

Perceived Income Risk Across the Income Distribution

Abstract

Low-income workers report wage uncertainty about twice as large as high-income workers, conditional on staying in the same job, yet matched administrative spells show no comparable gradient in realised risk, so the gap is one of beliefs. I show that a single learning rule organises three features of perceived risk by income: its level, its persistence, and its response to realised shocks. Estimated within person, the bottom updates most slowly, reads, in point estimates, the largest share of each shock as permanent, and carries a prior far above its realised transitory risk. I trace the gradient to how pay is delivered. Perceived risk departs from realised risk in opposite directions across the distribution, the bottom over-perceiving its transitory risk and the middle and the top under-perceiving their permanent risk, a two-signed pattern no friction of a single sign can produce. The gradient is one of how legibly pay separates into permanent and transitory parts. At the top, transitory pay is labelled and netted out before the update, a share I measure directly from the itemised pay components of the same administrative records, rising from about an eighth of transitory pay variance at the bottom to over a third at the top; the measured share sets the persistence, the response to shocks, and the under-perception higher up, and its netting floors predict, within a standard error, the level at which the middle and top priors sit. The bottom's over-perception is a second force of the opposite sign that separation cannot deliver, a noisier reading of fragmented pay identified as the residual its prior holds above the measured floor. Embedded in a life-cycle buffer-stock economy, the measured beliefs open a precautionary-saving and marginal-propensity-to-consume wedge, rising with income to six percent of liquid wealth at the top under the measured shares, eight at the upper edge of the measurement band, and sixteen where the estimated belief responses are taken at face value, that a full-information calibration cannot generate; once workers drift across income groups the behavioural wedge concentrates at the top, which under-perceives the permanent risk that drives saving most, and spares the bottom, whose distortion sits in a transitory prior that saving barely prices.

1 Introduction

In an incomplete-market economy, how much risk a household believes it faces is a primitive of its consumption and saving behaviour. Holding expected income and preferences fixed, an agent who perceives more risk to her future earnings accumulates a larger precautionary buffer and spends less out of a windfall, whether because prudence in the utility function or an occasionally binding borrowing constraint induces precautionary saving. The size and the heterogeneity of idiosyncratic income risk are accordingly among the central inputs of the heterogeneous-agent models now standard in macroeconomics, where they shape the distribution of wealth, the distribution of marginal propensities to consume, and, through both, the transmission of aggregate shocks and of policy. But is the risk these models assign their agents the risk those agents actually perceive, and if not, does the gap between the two fall evenly across households or pull them apart?

The common practice in this literature is to calibrate income risk indirectly. The econometrician estimates the cross-sectional dispersion of realised earnings in panel data, decomposes it into components of differing persistence (Meghir and Pistaferri, 2004), and hands the resulting variances to the agent as the risk she perceives. Perceived risk is set equal to realised risk: the agent is assumed to know exactly the variances the econometrician recovers, and to know nothing the econometrician cannot. The assumption is convenient but strong, and direct data on perceptions now allow it to be tested.

Measured directly, perceived risk violates this assumption, and it does so systematically across the income distribution. In the New York Fed’s Survey of Consumer Expectations (SCE), which elicits from each respondent a density forecast of her own wage growth over the next twelve months, workers in households earning under \$50k report wage uncertainty about 1.8 times as large as workers in households above \$100k, conditional on staying in the same job. Their beliefs are also more persistent within person, and they revise them more strongly after a realised wage shock. None of these three features appears in the realised wage process: matched same-job spells in the Survey of Income and Program Participation (SIPP) show no comparable gradient in realised risk, and if anything realised risk rises mildly with income. The gradient is therefore one of beliefs, not of realised risk, and the practice of equating the two fails on all three margins at once, on the level, on the persistence, and on the response to shocks.

This paper documents these patterns, fits a learning rule that reproduces them, and traces them to how pay is delivered across the income distribution, in three steps. The first is descriptive: after Section 2 sets out the measurement, Sections 3 and 4 establish the belief gradient and its dynamics against a realised process that shows no counterpart, fixing what the rule must reproduce.

The second step fits a belief-updating rule. The agent does not observe her permanent and transitory shocks separately; she sees one squared pay change each period and must learn the variance of her permanent earning power from it. In Section 5 I model this as an exponentially weighted average of past squared surprises, which is the steady-state Kalman filter for an unknown variance modelled as a random walk. The rule has three interpretable parameters: how fast beliefs update, what share of each squared surprise the agent reads as informative about permanent earning power, and a prior on transitory risk. Section 6 takes the rule to the data and reports the estimates by income group: estimated within person, the bottom updates most slowly, attributes to permanent earning power essentially the share a full-information agent would while the middle and the top attribute about half of theirs, and holds a prior almost four times its realised transitory risk, while the middle and the top sit close to one another and near their realised processes. The persistence and the prior gradients are statistically significant; the attribution gradient is monotone but not.

The third step, in Section 7, asks why the deviations take the form they do, and here the paper departs from the standard model. The fitted rule reproduces the persistence and the level of the gradient, but it does not explain them, since its three parameters are estimated freely group by group. What they leave unexplained runs in two directions at once: measured against the risk each group actually faces, the bottom *over*-perceives its transitory risk while the middle and the top *under*-perceive their permanent risk, in the same cross-section, and a single one-sided friction cannot produce a gradient that crosses its own benchmark in this way. I propose a mechanism that can, and measure its primitive: a worker forms her risk belief from a pay change she must first

separate into a permanent and a transitory part, and how cleanly she can separate it differs across the distribution. The data therefore require one force of each sign: the netting of the pay a worker can label pushes her perceived risk down, and the residual her prior holds above the measured netting floor, an excess of about $7.4(\%)^2$ at the bottom against residuals indistinguishable from zero above it, is the upward force, separately identified once the floor is measured (Appendix F).

The mechanism turns on how legibly pay separates into permanent and transitory parts, graded by income. At the top transitory pay arrives labelled and from a single salary, so the worker nets it out before she updates and her perceived risk falls; at the bottom it arrives fragmented and in ambiguous shocks, so the transitory part cannot be separated and her perceived risk is inflated. The labelled share is not a free parameter: the SIPP itemises the components of pay, and on the same stayer panel that supplies the realised process the share of transitory pay variance that arrives labelled rises from about an eighth at the bottom to over a third at the top. With the share measured, the estimated belief moments become predictions to be scored rather than moments to be fitted, and the measured netting floors predict, within a standard error, the level at which the middle and top priors sit. This separation sets the persistence, the response to news, and the under-perception higher up; the bottom's over-perception is the noisier reading of pay it cannot separate. The income gradient in beliefs is then a gradient in how pay is delivered, not in how risky pay is: legible pay is at once cleaner information and a sharper signal the worker under-reads, and delivery grades both.

What the estimated mechanism implies for behaviour follows analytically, before any simulation. In Section 8 I show that the gap between an agent's perceived second moments and the true ones is a perceived-risk wedge, and because it is an error about a variance rather than a level, it acts on saving through prudence rather than through the intertemporal-substitution motive. It therefore bears on saving through the perceived risk to permanent earning power, and hardly at all through the transitory prior, which a buffer absorbs within the period. In Sections 9 and 10 I embed the measured beliefs in a life-cycle buffer-stock economy in which workers drift across income groups as their permanent earning power accumulates, and in which a single discount factor, common across agents, is calibrated to average liquid wealth, so that the cross-section of wealth and marginal propensities is an outcome of the risk process. Relative to an otherwise identical full-information calibration, the beliefs open a precautionary-saving and marginal-propensity-to-consume (MPC) wedge that the full-information calibration cannot itself generate. The saving wedge rises with income, from under two percent of liquid wealth at the bottom to six percent at the top under the measured shares, eight at the upper edge of the measurement band and sixteen where the estimated belief responses are taken at face value, and the behavioural wedge in the MPC concentrates at the top, whose permanent belief is discounted most; liquidity reinforces the incidence but does not create it. In one-asset general equilibrium the belief economy supplies about three percent less capital and clears at an interest rate 0.12 percentage points higher, the wedge living in the cross-section more than in the aggregates. The bottom holds the most distorted transitory prior of any group, but its permanent belief is close to the truth, so the wedge spares it.

The mechanism of Section 7 is measured at its primitive and scored against belief moments it does not fit; its sharpest test remains out of sample, in variation in how pay is delivered. The model of Sections 9–10 is solved in partial and in one-asset general equilibrium; the quantified risk-shock

impulse responses and a two-asset general-equilibrium closure, the credible setting for the wedge’s effect on marginal propensities, are left to future work.

Related literature

The paper closest to this one is Wang (2023), who first used the SCE’s density forecasts to show that perceived wage risk lies well below the risk conventionally calibrated from panel data, even within demographic groups, and who showed that feeding this lower and more heterogeneous perceived risk into an incomplete-market model raises wealth inequality and the share of hand-to-mouth households, with the saving response operating mainly through an ex-ante “choice” channel rather than through realised outcomes. I share his measurement strategy, the SCE density forecast for perceived risk and the SIPP same-job stayer panel for the realised process, and I find the same sign for the middle and the top: at the twelve-month horizon they perceive less risk than the full-information benchmark implies, a gap whose size Section 4 reads as an upper bound, while the bottom sits essentially at the benchmark, its inflated transitory prior offsetting a mild permanent under-perception. I depart from him in four respects. First, I organise the deviation by income and show that, once decomposed, it is two-signed, the bottom over-perceiving its transitory risk and the middle and top under-perceiving their permanent risk, so the cross-section crosses its own benchmark rather than lying uniformly below it. Second, I document the within-person dynamics of perceived risk, its persistence and its response to realised shocks, where his evidence is cross-sectional. Third, I estimate an explicit updating rule, where he remains agnostic about the formation of beliefs by design. Fourth, I give the gradient a mechanism in how pay is delivered, measuring its primitive, the labelled share of transitory pay, from the itemised pay components of the same records and testing its predictions against the belief estimates, where he attributes the level gap to unobserved heterogeneity or to overconfidence without taking a stand on its source. The behavioural incidence differs accordingly: his account emphasises the bottom and aggregate inequality, while the wedge here concentrates at the top, because the saving-relevant distortion, the under-perceived permanent risk, is itself graded by income and largest there.

The paper connects to several further strands. On subjective beliefs about income, Rozsypal and Schlafmann (2023) document an income-graded bias in expected income *levels*; I study the second moment rather than the first. Caplin et al. (2023), contemporaneously and in Danish register-linked data, also find survey-reported earnings risk below its indirectly estimated counterpart, consistent with the under-perception I find higher up the distribution. The mechanism speaks to the long-standing question of insurance versus information in the partial-insurance literature: Kaufmann and Pistaferri (2009) show that a worker’s superior information can rationalise part of the gap between perceived and econometric risk, and the labelling of transitory pay here is exactly that channel, made observable and graded by income; the under-reaction that lowers perceived permanent risk and opens the wedge is a separate force, which I keep distinct. The learning rule builds on Pischke (1995), who casts the household’s inference about its permanent income as a signal-extraction problem, and the heterogeneous-information theme connects to Broer et al. (2022), who study information heterogeneity in heterogeneous-agent settings. Finally, the mechanism rests on documented facts about how pay is delivered across the distribution: performance pay concentrates at the top (Lemieux et al., 2009), high-income shocks are predominantly transitory (Guvenen et al.,

2021), bottom-of-distribution income is dominated by volatile and fragmented hours and gig flows (Hardy and Ziliak, 2014; Morduch and Schneider, 2017), and recessions fall hardest on the bottom (Hoynes et al., 2012).

2 Data and measurement

The paper needs two objects that can be set against each other: what a worker believes about her wage risk, and the risk her wage actually carries. I measure the first from the New York Fed’s Survey of Consumer Expectations (SCE) and the second from the Survey of Income and Program Participation (SIPP), both monthly within-respondent panels, the SIPP running over reference years 2013–2023 and the SCE over 2013–2025. I hold the two to the same conditioning, a worker who stays in the same job with the same employer for at least six consecutive months, so that any gap between them is a gap in beliefs and not in what is being measured, and I apply survey weights throughout. Respondents are sorted into three groups by median annual household income, the bottom below \$50k, the middle between \$50k and \$100k, and the top at or above \$100k.¹

The perceived object comes from the SCE density question, which asks each respondent for the distribution of her own earnings growth over the next twelve months “*on the same job, same hours, same employer*”, reported as a ten-bin histogram. The conditioning is what makes the answer usable: by fixing the job, the hours, and the employer, it strips out the predictable path of pay and the part a worker varies herself, so the elicited object is the conditional variance of the stochastic component of the wage change rather than of income, the quantity Section 4 sets against the realised process. Two consequences follow: the measure is free of the voluntary labour-supply variation that confounds risk estimates built on total earnings, and it excludes the risk of losing or changing the job, the larger part of income risk, which the model reintroduces separately through its employment block (Section 9). From each reported histogram I take the bin-midpoint variance as the perceived variance $\bar{\sigma}_{\text{perc}}^2$, exclude respondents who place all their mass in a single bin, trim at 1%/99%, and drop each respondent’s reports from the first wave in which she records a job change.

The realised object comes from the SIPP, which I use in preference to the Panel Study of Income Dynamics because it carries a unique employer identifier and records earnings every month, so a worker can be held to the same job and employer at the frequency the SCE question asks about, where the biennial PSID cannot. Per-month earnings are regular wage and salary together with bonuses, commissions, tips, and overtime, the amounts actually paid, and the self-employed are excluded. I drop the January seam, where the reported cross-wave change bunches spuriously, and trim squared changes at 1%/99%.

I residualise both objects on the standard demographic controls before any moment is computed: age and its square, sex, education, state, and year-month fixed effects. This leaves the variation a worker’s position in the income distribution explains, net of what her demographics and location predict. The SCE residualises the perceived variance $\bar{\sigma}_{\text{perc}}^2$ and the SIPP the log wage; year-month effects absorb aggregate shocks common to all groups. The SCE alone carries a survey-tenure term

¹The cutoffs are nominal and fixed across the sample. The SCE skews toward higher incomes, so group sizes are uneven (estimation sample: 1,531 / 7,501 / 11,331 observations).

that absorbs the panel conditioning of repeated subjective reports (Kim and Binder, 2023), absent from realised pay.

3 Perceived risk by income group

Levels

Perceived variance is steeply graded by income, with the bottom of the distribution far above a middle and top that sit together. The weighted group means are 20.6 at the bottom, 11.3 in the middle, and 11.4 at the top, so a worker at the bottom perceives about 1.8 times the wage uncertainty of one at the top, and Figure 1 rejects the equality of the two. The gradient is a step rather than a slope: the middle and the top are statistically indistinguishable and track each other closely, while the bottom sits some nine points above both. The bottom series also drifts upward over the sample, which may reflect the fixed nominal cutoffs or the greater cyclical exposure of low-income pay; Appendix D.2 shows the estimated belief block is insensitive to it. The bottom cut-off is set at \$50k because the finer income brackets are too thin to support the estimation (Appendix F); a finer split would have located more precisely where in the distribution the gradient forms. Part of the level gradient reflects who sits at the bottom: the SCE’s numeracy module places 47 percent of bottom-group respondents in its low-numeracy category, against 28 and 11 percent above, and within high-numeracy respondents the bottom-to-top gap compresses to roughly a quarter of its full-sample size yet remains significant ($p = 0.002$; Appendix D.4). I read the conditioned gradient as a lower bound rather than the cleaner estimate, since a coarse reading of one’s own pay is part of the mechanism itself, so conditioning on numeracy partials out part of the channel, the same argument that keeps occupation out of the baseline.

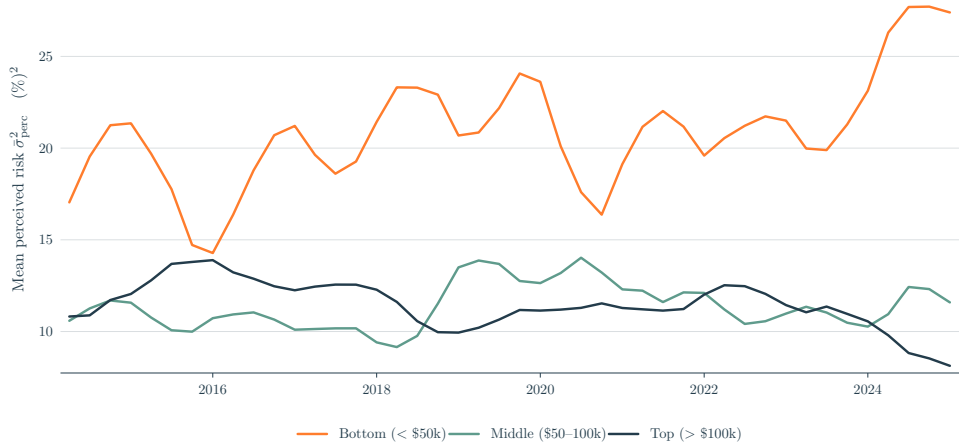


Figure 1: Group-mean perceived variance $\bar{\sigma}_{\text{perc}}^2$ in $(\%)^2$, quarterly with a 4-quarter trailing MA. Matched-stayer SCE panel, residualised as in Section 2.

Persistence

The level says nothing about how perceived risk moves, and its dynamics are graded as well. I estimate a within-respondent AR(1) in the perceived variance $\bar{\sigma}_{\text{perc}}^2$, with persistence coefficient $\hat{\phi}_g$, by Blundell-Bond system GMM (Blundell and Bond, 1998). The short, wide SCE panel makes

the within estimator severely Nickell-biased, so I instrument the lagged level with deeper lags to remove that bias, following Ma et al. (2020), who opts for this estimator for the same short-panel reason.

Group	Monthly		Quarterly	
	$\hat{\phi}_g$	SE	$\hat{\phi}_g$	SE
Bot	+0.343***	0.046	+0.632***	0.062
Mid	+0.142***	0.026	+0.594***	0.088
Top	+0.144***	0.025	+0.377***	0.076
Wald ($B = T$)	$p = 0.0001$		$p = 0.009$	

Table 1: Within-respondent AR(1) persistence $\hat{\phi}_g$ of perceived variance, Blundell-Bond system GMM. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 in differences plus the BB level moment, collapsed. Two-way clustered SEs (respondent, quarter). The same instrument set is used at both cadences.

The bottom is the most persistent of the three groups, and its equality with the top is rejected at both cadences. The gradient is also not inherited from the realised process, whose squared changes are if anything less persistent at the bottom (Appendix D.5). The quarterly estimates exceed the monthly ones for every group, which the time aggregation of an AR(1) cannot produce; classical noise in the monthly reports cannot fully produce it either, since the attenuation required would imply an implausibly persistent true process, so I read the two cadences as bracketing the true persistence and rest on what they agree on, the cross-group ordering. Taken with the level, the persistence leaves the standard heterogeneous-agent assumption that perceived risk equals realised risk with two gradients to reproduce, not one: a calibration from realised wage data would have to match both the level and the persistence documented here.

4 The realised wage process

If the belief gradient reflected a gradient in actual risk, it would show up in the realised wage process, and it does not. Realised shocks nonetheless move beliefs, and together these two facts pin down what a belief rule must deliver.

Wage model

I use the standard permanent-transitory decomposition (Meghir and Pistaferri, 2004):

$$w_{i,t} = z_{i,t} + e_{i,t}, \quad e_{i,t} = p_{i,t} + \theta_{i,t}, \quad p_{i,t} = p_{i,t-1} + \psi_{i,t}.$$

Here $w_{i,t}$ is the residualised log wage, $z_{i,t}$ the predictable Mincerian path, and $e_{i,t}$ the stochastic component that is the focus of the rest of the paper. The permanent shock $\psi_{i,t}$ is iid with variance σ_ψ^2 and accumulates through the random walk $p_{i,t}$ (a merit raise or cost-of-living adjustment); the transitory shock $\theta_{i,t}$ is iid with variance σ_θ^2 (a discretionary bonus or profit share).

Matching SCE to the model

The SCE question conditions on *same job, hours, employer*, which fixes $\Delta z = 0$, so the elicited variance is that of the stochastic component alone:

$$\bar{\sigma}_{\text{perc},i,t}^2 \equiv \text{Var}_{i,t}(\Delta w_{i,t+1} \mid \Delta z = 0) = \text{Var}_{i,t}(\Delta e_{i,t+1}).$$

On SIPP I residualise (removing z) and difference to recover Δe directly. The recorded wage is the amount paid per usual hour, so it includes tips, overtime, and any gap between actual and usual hours: the recorded counterpart of the worker’s paycheck. Both sides therefore measure $\text{Var}(\Delta e)$.

Identification of the wage primitives

The one-period change is $\Delta e_t = \psi_t + \theta_t - \theta_{t-1}$. Write $\bar{S}_m$ for the variance of this monthly change, $\mathbb{E}[(\Delta e)^2]$. Because ψ and θ are independent and iid, it has only two non-trivial moments per group:

$$\bar{S}_m = \sigma_\psi^2 + 2\sigma_\theta^2, \quad \text{Cov}(\Delta e_t, \Delta e_{t-1}) = -\sigma_\theta^2.$$

Group	σ_θ^2 (transitory)	σ_ψ^2 (permanent)	$\bar{S}_m$
Bot	2.52	11.92	16.96
Mid	4.42	11.61	20.45
Top	5.83	11.23	22.89

Table 2: SIPP permanent-transitory identification per income group, monthly cadence, in $(\%)^2$. Broad stayer panel, January seam dropped.

The lag-1 autocovariance identifies σ_θ^2 and the remaining variance identifies σ_ψ^2 , so the two wage primitives are exactly identified per group. Permanent variance is roughly flat across groups, while transitory variance rises mildly with income, plausibly a bonus channel at the top. The realised gradient therefore runs, if anything, *opposite* to the perceived one.

The FIRE benchmark

To compare the monthly process with the 12-month SCE horizon, I roll it forward twelve months under iid shocks (derivation in Appendix B):

$$\text{Var}(e_{t+12} - e_t) = 12\sigma_\psi^2 + 2\sigma_\theta^2, \tag{1}$$

which is what a full-information rational-expectations (FIRE) agent who knows the parameters would report. The benchmark is the conventionally calibrated risk, near $148(\%)^2$ for every group, and perceived variance falls far below it, to 20.6, 11.3, and 11.4, between a seventh and a thirteenth of the benchmark (Table 6, Appendix B). That gap is the one Wang (2023) documents, and I read it cautiously, resting nothing on the absolute ratios and everything on the cross-group contrast. The ratios overstate miscalibration for a reason mechanical in the benchmark: equation (1) scales permanent variance by twelve but transitory by only two, so any contamination loading onto σ_ψ^2 is magnified twelvefold. SIPP earnings are self-reported, and reporting error is exactly such

a contaminant, its classical component differencing into an MA(1) read as transitory while any persistent component is indistinguishable from a true permanent shock and is then multiplied by twelve (Bound and Krueger, 1991); administrative payroll data place job-stayers' base-wage dispersion well below the benchmark for the same reason (Grigsby et al., 2021). That error, however, is common across the groups and cancels in the cross-group comparison, so it cannot sign the contrast. Section 9 returns to this benchmark in SIPP units on both sides, where the contamination cancels regime against regime.

Realised shocks still move beliefs

The cross-section rules out the realised process as the *source* of the gradient, but the realised process can still drive belief *dynamics*. Figure 2 shows that pooled perceived risk co-moves with the absolute deviation of aggregate wage growth from its sample mean, consistent with surprises to wage growth moving perceived risk over time.²

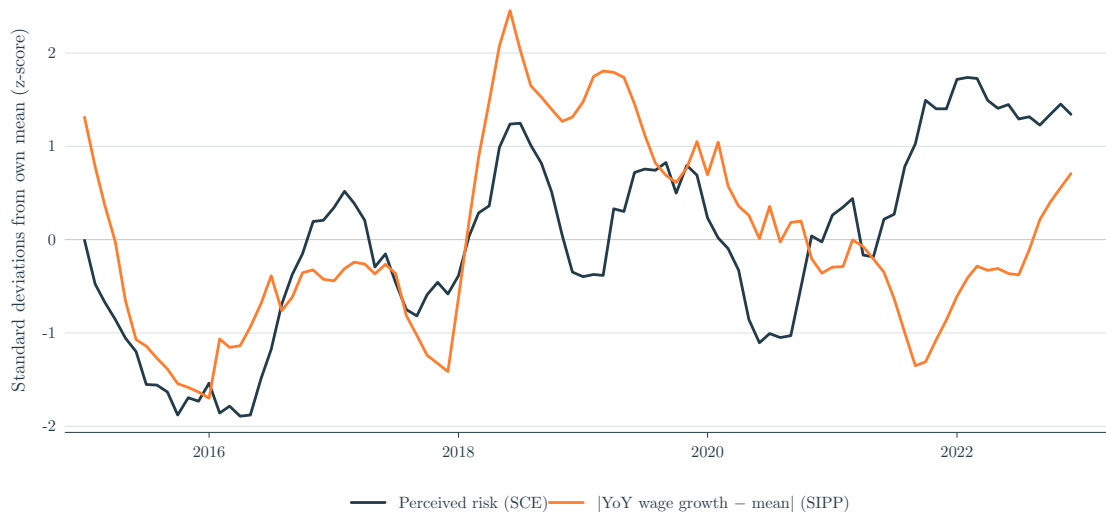


Figure 2: SCE perceived risk and the absolute deviation of SIPP aggregate wage growth from its sample mean, both standardised, pooled across income groups, 2015–2022.

Together these facts impose two requirements on any belief rule. It must let perceived variance move, since the dynamics rule out constant beliefs; and it must let realised shocks drive that movement, with the response to the largely common signal differing by group.

5 A belief-updating rule

A reasonable assumption I will make is that an agent cannot tell a permanent shock from a transitory one. Each period she sees a single squared pay change, which mixes the two, since $\mathbb{E}[(\Delta e)^2] = \sigma_\psi^2 + 2\sigma_\theta^2$, and she must split that surprise, reading a share λ as news about her permanent earning power and the rest as transitory. The SCE gives one perceived-variance series per group, enough to pin down one moving belief but not two, so I let the permanent belief

²Absolute deviations, since any realised shock should raise perceived uncertainty; squared deviations give a very similar plot.

$\hat{\sigma}_{\psi,t}^2$ evolve and hold the transitory belief at a static prior $\sigma_{\theta,\text{perc}}^2$. The permanent component is the welfare-relevant one and the natural place for the dynamics; the price is that any belief-side misattribution loads onto λ . The allocation is an identifying normalisation rather than a finding: with one series per group, the constant could equally be read as a static permanent bias with the dynamics in the transitory belief. The differenced reading error of Section 7 (eq. (5)) carries the variance signature of a transitory shock, variance $2\omega^2$ and first autocovariance $-\omega^2$, which makes the prior its natural home; under the alternative allocation the bottom's wedge would turn into modest over-saving while the top's under-saving, pinned by the netting channel, survives.

The permanent-variance belief follows an exponentially weighted moving average (EWMA) of past squared shocks, with the attribution share as a separate weight:

$$\hat{\sigma}_{\psi,t}^2 = \underbrace{(1-\mu)}_{\text{persistence}} \hat{\sigma}_{\psi,t-1}^2 + \underbrace{\mu}_{\text{update weight}} \underbrace{\lambda}_{\text{attribution share}} (\Delta e_{t-1})^2. \quad (2)$$

Here μ is the update speed, a higher μ moving the belief more on each period's news, and λ the share of each squared surprise read as informative about permanent earning power. Because $(1-\mu)$ and μ sum to one, the recursion is the integrated GARCH(1,1) in the squared surprise; it is equally the steady-state Kalman filter for an unknown permanent variance modelled as a random walk, with μ the Kalman gain (Appendix F gives the filter and sets the gain by the signal-to-noise ratio). Given this information structure the agent applies the steady-state Kalman gain each period, the perpetual-learning rule for a drifting variance; what kind of departure from full information her constant attribution then represents, once legibility grades it, is taken up in Section 7.

Substituting the belief into the FIRE identity (1) gives the level the agent reports:

$$\bar{\sigma}_{\text{perc},t}^2 = 12 \hat{\sigma}_{\psi,t}^2 + 2\sigma_{\theta,\text{perc}}^2. \quad (3)$$

Strict FIRE is the corner $\mu = 0$ with both beliefs at the truth, a constant reported variance that the dynamics of Section 3 rule out. Within the learning family ($\mu > 0$) FIRE survives only as a long-run average: taking unconditional expectations of (2) in a stationary state gives $\mathbb{E}[\hat{\sigma}_{\psi}^2] = \lambda \bar{S}_m$, so the belief is right on average exactly when $\lambda^* = \sigma_{\psi}^2 / \bar{S}_m$, which Table 2 puts at 0.70, 0.57, and 0.49 across the three groups, with the transitory condition $\sigma_{\theta,\text{perc}}^2 = \sigma_{\theta}^2$ alongside. These two conditions are what the estimates test.

6 Estimation

The belief recursion (2) is written in the unobserved $\hat{\sigma}_{\psi}^2$. Combining it with the level identity (3) eliminates the belief and leaves an AR(1) in the *observable* perceived variance (Appendix A):

$$\bar{\sigma}_{\text{perc},i,t}^2 = \underbrace{2\mu_g \sigma_{\theta,\text{perc},g}^2}_{\beta_{\text{cons}}} + \underbrace{(1-\mu_g)}_{\beta_{\text{lag}}} \bar{\sigma}_{\text{perc},i,t-1}^2 + \underbrace{12\mu_g \lambda_g}_{\beta_{\text{shock}}} (\Delta e_{i,t-1})^2 + \alpha_i + u_{i,t}, \quad (4)$$

with α_i a person fixed effect. The three coefficients map one-to-one to the structural parameters: $\mu_g = 1 - \hat{\beta}_{\text{lag}}$, $\sigma_{\theta,\text{perc},g}^2 = \hat{\beta}_{\text{cons}} / (2\mu_g)$, and $\lambda_g = \hat{\beta}_{\text{shock}} / (12\mu_g)$. Each is identified from a distinct feature of the data: β_{lag} from the within-respondent persistence, β_{cons} from the group level once

the news loading is netted out, and β_{shock} from the response to the lagged realised shock.

The shock regressor

The rule calls for the agent’s own $(\Delta e_{i,t-1})^2$. The SCE contains no individual wage histories, so I substitute the SIPP cell mean over workers in the same cell $c = (\text{group} \times \text{Census division} \times \text{month})$, following Guerreiro (2023); workers in the same local labour market share components of wage news, transmitted through coworkers and local media and salient for expectations (Kuchler and Zafar, 2019). The cell series is smoothed with a 3-month trailing MA.

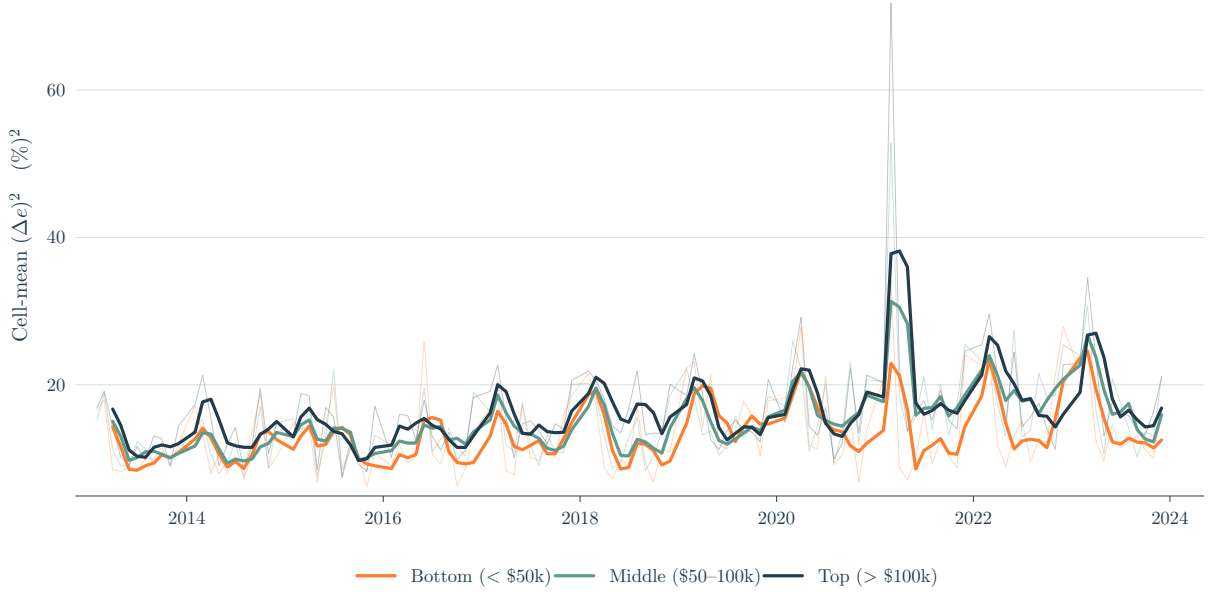


Figure 3: Cell-mean $(\Delta e_{i,t})^2$ by income group, national monthly, in $(\%)^2$. Thin lines: raw monthly cell means. Thick lines: the 3-month trailing MA used as the regressor in (4).

The top has the largest shock amplitude, consistent with its higher σ_θ^2 and with discretionary pay concentrating at the top (Güvenen et al., 2021; Halvorsen et al., 2024). Because the bottom’s shocks are no larger than the top’s, any stronger belief response at the bottom is a property of beliefs, not of shocks.

Replacing the own shock with a cell mean is a measurement-error problem that attenuates β_{shock} , and hence $\hat{\lambda}$, toward zero. The attenuation works through finite-cell sampling noise: the cell mean tracks only the common component of a worker’s shock, so its informativeness about her own shock depends on how large the common component is and on the cell size. The level of $\hat{\lambda}$ is therefore not interpretable on its own, and what follows reads its cross-group pattern rather than its level.

I instrument with $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4 and the shock at lags 1–3, together with BB level moments. The shock is treated as endogenous: the contemporaneous shock and the belief share a survey-timing and aggregate-news channel, and the Hansen J test rejects the contemporaneous moment set but does not reject lags 1–3.

Estimates by income group

Joint GMM on (4) by income group yields the estimates in Figure 4, set against the realised SIPP transitory variance. The full estimates, with standard errors, the Hansen J , and sample sizes, are in Table 7 (Appendix D).

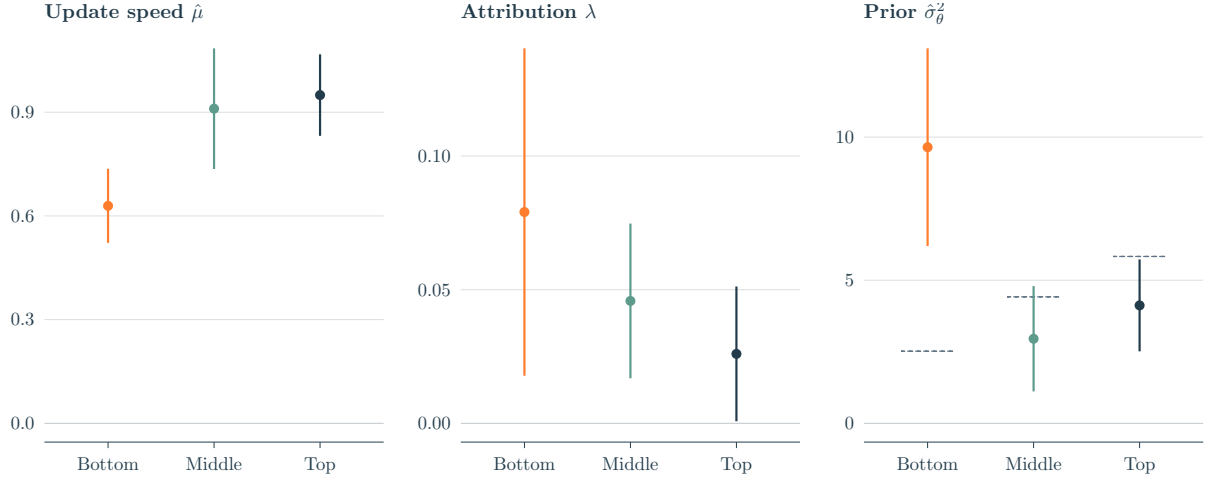


Figure 4: Headline estimates by income group: update speed μ_g , attribution λ_g , and transitory prior $\sigma_{\theta, \text{perc}, g}^2$, with 95% confidence intervals. Grey dashes in the prior panel mark realised transitory variance σ_θ^2 (SIPP). The attribution is identified in cross-group ratios, so its panel shows slope, not level.

Update speed. The bottom updates significantly more slowly than the middle or top, matching the persistence ordering of Table 1. The influence of a given wage shock on its belief therefore decays more slowly.

Attribution. Each $\hat{\lambda}_g$ is positive and significant, and the point estimates fall monotonically with income, the bottom's being three times the top's, although cross-group equality is not rejected. A common wage signal generates larger revisions of the permanent belief at the bottom, in point estimates if not yet in significance. Part of the decline is itself rational: the full-information share λ^* falls from 0.70 to 0.49 across the groups (Section 5), a gradient of about 1.4, so it is the excess of the estimated three-to-one slope over that benchmark that is belief content.

Transitory prior. Realised transitory variance rises with income; the perceived prior does the opposite. The bottom's prior sits almost four times its realised value, while the middle and top lie near or below theirs. The bottom is the only group whose prior is materially detached from its realised data, and the gradient is significant. The inflated prior is the dominant driver of the *level* of the bottom's perceived variance, separate from the news channel.

Robustness

Adding an occupation $\times$ month fixed effect to the shock construction preserves the persistence and attribution gradients and sharpens the latter, while the prior gradient weakens (Appendix D.1); I do not adopt it as the baseline, since low-income workers sort into structurally more uncertain occupations (Morduch and Schneider, 2017), so partialling out occupation removes part of the channel. When the shock is split by sign and groups are pooled for power (Appendix D.3), both components load positively, but upside news carries about twice the weight of downside (0.055

versus 0.027). My interpretation is that workers read good news as informative about permanent earning power and dismiss bad news as transitory, although this is the opposite of what loss aversion would predict; a second and more cautious reading is that pooling the groups confounds their group-specific responses, so the asymmetry cannot be read reliably. Appendix D.2 varies the instrument lags, the trim, and the COVID handling one at a time: the persistence gradient survives every variant except a tighter within-group trim, and the attribution gradient stays monotone throughout.

7 A pay-delivery mechanism

The estimates of Section 6 give three parameters for each income group, the update speed, the attribution, and the transitory prior, each fitted freely, so the gradients across groups are documented but left unconnected. This section asks what single primitive could generate them together. I propose that it is a single feature of how pay is delivered: how legibly a worker’s pay separates into a permanent and a transitory part. Legibility is graded by income, and that grading sets the persistence of beliefs, their response to news, and the amount by which they fall short of the truth. It does not set everything: the bottom’s prior needs a second force, taken up below. The primitive is not a conjecture to be calibrated. The SIPP itemises the components of pay, so the share of transitory pay that arrives labelled can be measured on the same stayer panel that supplies the realised process, and once it is measured the mechanism carries no free parameter: the estimated belief gradients become predictions to be scored against it, not moments to be fitted. Section 9 carries the measured primitive into the model.

The legibility of pay, and the bottom’s prior

What the mechanism must explain is the shape of the transitory prior, since it is there that perceived risk crosses its own realised benchmark. The estimated prior sits far above realised transitory variance at the bottom, 9.65 against 2.52, a gap of nearly four to one that no plausible units or reporting-error correction closes, and at or below it at the middle and the top, 2.96 against 4.42 and 4.12 against 5.83, shortfalls of one to two standard errors that I read as directional only, since the two surveys’ levels are compared here at face value (Section 4). A single force that pushed perceived risk to one side of the truth could not produce both ends; two forces can, provided each acts in only one direction. Separating the transitory part of a worker’s pay and netting it out before she updates can only lower her prior, while pay too fragmented to read cleanly can only raise it, since an unresolved reading carries the variance signature of a transitory shock and is recorded as one. The income gradient in the prior is then a question of which force dominates where, and both forces are features of how pay is delivered: the downward force is the share of transitory pay that arrives labelled, which the SIPP itemises and I measure directly below, and the upward force is the residual that the measured share leaves the prior sitting above.

Legibility and the labelling of transitory pay. Take the downward force first. A salaried worker at the top receives her transitory pay already flagged: a bonus or a commission arrives on its own line, a furlough comes with a return date, and the whole of her pay is a single salary to read against. A worker at the bottom receives the same kind of pay scattered and unlabelled, as

variable shifts, cash tips, and supplements layered on uncertain hours, and her downturns arrive ambiguous, a sector closing or gig work drying up rather than a furlough that will end. The first worker can tell which part of a pay change is transitory and net it out before she updates; the second cannot. Three features of delivery grade this legibility by income. Lemieux et al. (2009) document that performance pay arrives on separate line items and concentrates at the top; Hoynes et al. (2012) that high-income downturns arrive classifiably where low-income ones do not; and Hardy and Ziliak (2014) and Morduch and Schneider (2017) that high-income pay is a single salary where low-income pay is fragmented. Let $\rho_g \in [0, 1]$ be the share of transitory pay a worker can separate and net out. The separated share is removed before she updates, so only the unlabelled residual feeds the belief recursion (2), and her prior falls to the netting floor $(1 - \rho_g)\sigma_\theta^2$.

Measuring the labelled share. The share can be measured, because the SIPP records not only total earnings but the monthly dollar amounts of their components, bonuses, commissions, tips, and overtime pay, on the same same-job stayer panel of Section 2. I demean each component within a job spell, scale the within-spell dollar variation by spell-mean earnings, and take ρ_g as the share of within-spell transitory variance that the labelled components, bonuses and commissions in the main specification, account for, weighting throughout by the survey weights. On 2.25 million stayer months across 103,134 workers and 127,184 job spells over reference years 2013–2023, the labelled share is 0.121 at the bottom, 0.178 in the middle, and 0.358 at the top, rising with income as the delivery evidence above predicts. The gradient survives every variation I try, sitting between an upper edge that detrends rather than demeans each spell (0.176/0.256/0.488) and a lower edge that drops months with imputed component amounts (0.042/0.085/0.314); Appendix F reports the full specification table. At the measured shares the netting floors $(1 - \rho_g)\sigma_\theta^2$ are 2.21/3.63/3.74.

Reading noise and the bottom's prior. The upward force is the reading error that imperfect separation leaves behind. After netting out the labelled share, the worker still has to read the residual change, and when pay is least legible, variable shifts layered on uncertain hours, the reading itself is noisy: she sees the deposit, but she cannot cleanly tell a one-off supplement from a lasting raise behind it. Formally, her reading of the unlabelled part of her pay is

$$\tilde{y}_t = p_t + \theta_t^u + \nu_t, \quad \nu_t \stackrel{\text{iid}}{\sim} (0, \omega_g^2), \quad (5)$$

with p_t her permanent component, θ_t^u the unlabelled transitory component of variance $(1 - \rho_g)\sigma_\theta^2$, and ν_t the reading error, a signal-extraction structure in the style of Pischke (1995). The error lives only in the reading she feeds her risk inference, never in the resources she receives: the budget of Section 9 uses realised income, so ν touches no allocation directly.³ Recorded as risk, it lifts the prior above the netting floor:

$$\sigma_{\theta, \text{perc}, g}^2 = (1 - \rho_g)\sigma_\theta^2 + \omega_g^2, \quad \omega_g^2 \geq 0. \quad (6)$$

Netting can only lower the prior, so this is the one piece of the gradient the labelled share cannot deliver, but with the floor measured it is separately identified for the first time, as the residual of the

³Recorded SIPP earnings capture the amounts actually paid, so ν never enters the moments of Section 4. The differenced error $\Delta\nu_t$ has variance $2\omega_g^2$, first autocovariance $-\omega_g^2$, and none beyond, exactly the signature of a differenced transitory shock, which is why an unresolved reading is recorded as transitory risk, why the prior is its natural home (Section 5), and why it enters the reported level (3) with the same factor of two as the prior.

estimated prior above its floor: $\hat{\omega}^2 = 7.43$ at the bottom, with a standard error of 1.76 ($t \approx 4.2$; the prior's standard error, conditional on the measured floor), against residuals indistinguishable from zero at the middle and the top (-0.68 and $+0.38$, standard errors 0.94 and 0.82). It is a measured residual of the account, not a calibrated primitive: its level enters nothing downstream, since the model carries the prior itself, and only its non-negativity is imposed, as the floor-admissibility constraint of the reconciliation in Section 9, binding at the middle alone. At the bottom the identification does not depend on the share: the residual exceeds 7.1 at any $\rho \in [0, 1]$. The recursion (2) is written on the full change Δe ; equivalently the worker nets the labelled share and feeds the squared reading $(\Delta \tilde{y})^2$ into it with the gain rescaled so that the resting mean is unchanged (Appendix F), the form the household block of Section 9 carries. The residual carries the income gradient the account requires: within-person hours volatility, the natural source of unresolvable readings, falls monotonically with income (4.10/3.06/2.36; Appendix E), highest exactly where the residual is.

The measured floors do more than isolate the residual: they predict the middle and top priors out of sample. Netting implies a prior below realised transitory risk by exactly $\rho_g \sigma_\theta^2$, a shortfall of 0.79 in the middle and 2.09 at the top, against estimated shortfalls of 1.46 and 1.71, each within one standard error of its prediction ($z = -0.7$ and $+0.5$): the priors sit on their measured floors. The prediction extends to the one feature of the prior that looks like noise. The estimated priors are mildly non-monotone, the middle below the top, and the floors are non-monotone in exactly the same way, because realised transitory risk rises with income faster than the labelled share removes it; nothing has to be smoothed away. Because each force is one-sided, neither alone could produce a prior that crosses its benchmark; the measured share and the identified residual together produce precisely the crossing the estimates record.

From the measured share to the estimated gradients

With the labelled share measured, the remaining estimated parameters become predictions rather than targets. The single primitive ρ_g reaches each through one channel, since labelling shrinks the part of a pay change the worker must still interpret: it sets the attribution here and the permanent belief that drives saving in Section 8, and it orders the update speed. The gain is carried into the model as the estimated moment itself, the prior as the estimated moment wherever it clears its netting floor, and the attribution follows from ρ_g in closed form.

The transitory prior. The two forces meet in the prior, the sum (6): netting lowers it to its floor $(1 - \rho_g)\sigma_\theta^2$ and reading noise lifts it back above realised risk at the bottom, the net of the two the crossing in the prior panel of Figure 4. I carry the prior as the estimated moment wherever the estimate clears its netting floor, and at the floor where it sits below, since the mechanism admits no prior beneath the floor; the middle and top values are then their floors to within a standard error, and the bottom value the floor plus the identified residual.

The update speed. How fast a worker updates depends on how clean a signal each pay change is. Labelling removes part of the transitory noise, shrinking the unlabelled residual variance $V_u = \sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2$ relative to the whole change, so for a more legible worker each squared surprise is a sharper reading of her permanent risk and her steady-state Kalman gain is larger; the filter and the algebra are in Appendix F. The measured shares predict the ordering of the estimated gains

correctly, the top fastest and the bottom slowest, but underpredict their spread, 0.76/0.78/0.82 against the estimated 0.63/0.91/0.95 at the diagnostic drift of Appendix F. I weight the miss by the reliability of its target: the gain levels are the one set of belief moments the paper’s own cadence analysis discredits, the quarterly estimates exceeding the monthly ones in a way the time aggregation of an AR(1) cannot produce (Section 3), so the ordering rather than the spread is what the data pin, and the ordering is what the measurement delivers. The model accordingly carries the gains as the estimated moments.

The attribution share. The attribution falls with legibility for a related reason. A worker updates her permanent belief only on the unlabelled part of a pay change, but the regression of Section 6 has only the full change to work with, so the labelled part moves the regressor without moving the belief and attenuates the slope, exactly as classical measurement error in a regressor would. Under joint Gaussianity of ψ and θ this gives a closed form (Appendix C),

$$\lambda_g = \frac{\sigma_\psi^2(\sigma_\psi^2 + 2(1 - \rho_g)\sigma_\theta^2)}{(\sigma_\psi^2 + 2\sigma_\theta^2)^2}, \quad (7)$$

which falls in ρ_g and equals the FIRE benchmark $\lambda^* = \sigma_\psi^2/\bar{S}_m$ at $\rho_g = 0$. At the measured shares it is 0.677/0.524/0.401, declining with income as the λ panel of Figure 4 does. The comparable moments are the cross-group ratios, since the cell-mean shock regressor attenuates the estimated level by a factor common across groups (Appendix F): measured delivery predicts bottom-to-top and middle-to-top ratios of 1.69 and 1.31, against estimated ratios of 3.04 and 1.76 whose log standard errors are 0.63 and 0.59, each prediction within one standard error of its estimate and about half of the estimated point gradient. Delivery explains roughly half of how attribution declines with income; the remainder is either an under-reaction beyond what labelling accounts for or sampling noise in ratios this wide, a division the final subsection returns to.

From three groups to a profile. The model of Section 9 lets a worker’s income drift continuously, so it needs the belief objects as functions of income rather than three points; the share it carries is pinned within the measurement band above by the belief moments themselves, as the calibration of Section 9 describes. I interpolate the carried share and gain through the three group anchors with monotone cubic interpolants in levels, and the carried prior in logs, all held flat outside the anchor incomes with smooth joins, and the prior censored below at the local netting floor $(1 - \rho(y))\sigma_\theta^2(y)$ between them, since the mechanism admits no prior beneath it (Appendix F); the attribution and the permanent belief follow from the closed forms at each income. Nothing is smoothed: the prior’s dip at the middle survives in the carried block, since the measured floors predict it. Figure 5 draws the measured share, the identified reading noise, and the prior they compose against income, and Table 11 evaluates the mapping at representative incomes.

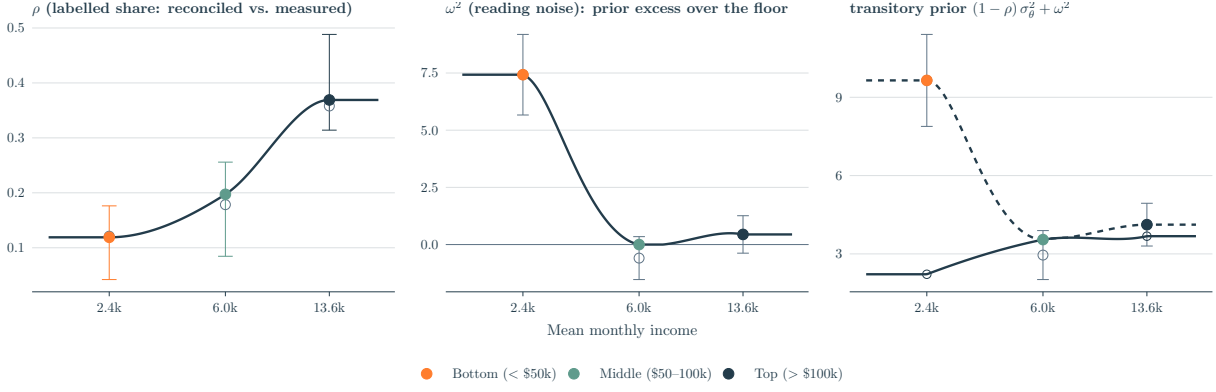


Figure 5: The labelled share ρ , the identified reading noise ω^2 , and the transitory prior they compose (6), by income. Left: the reconciled share the model carries (filled points, with its income interpolant), against the main-specification measurement (open points) and whiskers spanning the measurement-specification band of Appendix F, from the imputation-dropped lower edge to the detrended upper edge; the belief moments select the carried share within the band (Section 9). Middle: the reading noise, identified as the prior’s excess over its netting floor (the zero line), with the raw estimates’ excess and their ± 1 standard-error bands in grey; it is significant only at the bottom (7.4, $t = 4.2$). Right: the carried prior (filled points, dashed interpolant) as the sum of the netting floor $(1 - \rho)\sigma_\theta^2$ at the carried share (solid line, open ink points) and the reading noise, against the raw estimates and their ± 1 standard-error bands in grey; the middle is carried on its floor, and the bottom sits the identified 7.4 above it. In $(\%)^2$ except the dimensionless share.

Information and under-reaction

A last point concerns what kind of departure from full information this is, and it has two parts best kept apart. The first is genuine information. In netting out the share of transitory pay she can label, the worker removes risk she has correctly identified as transitory, and about that share she is unbiased: this is the superior-information channel of Kaufmann and Pistaferri (2009), a gap between perceived and econometric risk that reflects what the worker can see and the data cannot, and it is what legibility buys on the transitory side. The second is an under-reaction: the agent reads a fixed share of each surprise as permanent, and as netting turns that surprise into a cleaner signal she does not raise the share to match it, so her perceived permanent variance settles below the truth by a labelling discount, derived in Section 8, a downward deviation graded by legibility and of the kind familiar from constant-gain and experience-based learning. It is this second piece, not the first, that opens the saving wedge. The bottom labels little ($\rho_g = 0.12$), so it has little cleaner signal to under-read and its permanent belief is nearly unbiased, which is why the wedge largely spares it even though its transitory prior is the most distorted single belief of any group. The measurement also bounds how much of the estimated attribution gradient this under-reaction must carry: delivery predicts about half of the point gradient, so the residual half, if it is not sampling noise in ratios whose standard errors admit either reading, is an under-reaction over and above the constant gain, workers higher up discounting the permanent content of their pay changes by more than their measured labelling warrants. The mechanism therefore does not so much stand outside the behavioural accounts as run opposite to them: where diagnostic expectations add an over-reaction to news, the legible worker under-reacts to it, the downward half of the two-signed cross-section established above.

8 What the estimated mechanism implies

Section 7 measured a single pay-delivery primitive, the legibility ρ_g of a worker's pay, and carried the transitory prior alongside it as an estimated belief moment. In this section I work out what those two objects imply for behaviour, and I do so analytically, before the calibrated model of Section 9 attaches magnitudes to them. I proceed in three steps. I first write the perceived second moments in closed form, isolate the one piece of them that drives precautionary saving, and collect the comparative statics in legibility. I then define the gap between perceived and true risk that organises the rest of the paper, and show that, because it is an error about a variance and not about a level, it reaches behaviour through prudence rather than through the intertemporal-substitution motive. I finally let the belief carry its own dynamics, so that a shock to the level of income and a shock to its risk leave wedges of very different durability.

The permanent belief in closed form

Averaging the constant-gain recursion (2) over a stationary state cancels the persistence terms and leaves the mean permanent belief equal to the attribution share times the variance of the surprise the agent updates on, the full monthly change of Section 5: $\mathbb{E}[\hat{\sigma}_\psi^2] = \lambda \bar{S}_m$. The attribution share is itself graded by legibility: its closed form (7) is $\lambda = \sigma_\psi^2 V_u / \bar{S}_m^2$, the numerator's second factor being exactly the unlabelled residual variance V_u and the denominator $\bar{S}_m^2$. Substituting it, the belief settles at

$$\mathbb{E}[\hat{\sigma}_\psi^2] = \lambda \bar{S}_m = \frac{\sigma_\psi^2 V_u}{\bar{S}_m^2} \bar{S}_m = \sigma_\psi^2 \frac{V_u}{\bar{S}_m}. \quad (8)$$

The discount $V_u / \bar{S}_m \leq 1$ is the labelling discount. The same number can be read from either side of the netting: the agent books the attenuated share λ of the full change she observes, or, equivalently, she nets out the labelled part and updates on the residual of variance V_u while holding her attribution at the no-labelling benchmark $\lambda^* = \sigma_\psi^2 / \bar{S}_m$, since $\lambda^* V_u = \lambda \bar{S}_m$. Both readings count the netting a single time, and both say the same thing: the discount is the under-reaction of Section 7, the agent holding her attribution fixed as netting cleans her signal, so the more she nets the further her perceived permanent risk falls below the truth.⁴

This closed form is the long-run mean of Section 5, $\mathbb{E}[\hat{\sigma}_\psi^2] = \lambda \bar{S}_m$, evaluated at the measured labelling; the FIRE benchmark is its special case. The base is the same full monthly change in both; the two differ only in the share applied to it, the unbiased $\lambda^* = \sigma_\psi^2 / \bar{S}_m$ against the attenuated $\lambda \leq \lambda^*$ of (7). When nothing is labelled, $\rho_g = 0$ and $V_u = \bar{S}_m$, the two shares coincide and the belief is the unbiased σ_ψ^2 ; labelling is what drives them apart. One observational equivalence belongs on the record: a Bayesian who netted and raised her attribution to match the cleaner signal would rest at the unbiased σ_ψ^2 , and she generates the same slopes in the estimating equation (4), so the regression alone cannot reject a zero wedge. The level of the wedge rests on the maintained constant attribution of Section 5 together with the direct evidence that perceived risk sits below the benchmark higher up (Section 4; Caplin et al., 2023); what the measurement pins is its grading

⁴A third reading is the implemented one: the agent updates on her own noisy netted reading (5), whose variance is $V_u + 2\omega_g^2 = \sigma_\psi^2 + V_x$ with $V_x = 2\sigma_{\theta, \text{perc}, g}^2$, and the recursion's gain is the rescaled $\lambda^{\text{rule}} = \sigma_\psi^2 V_u / [\bar{S}_m(\sigma_\psi^2 + V_x)]$, chosen so that the resting mean is unchanged (Appendix F, Table 11): the reading error inflates her signal and the rescaling offsets it, so it never reaches the resting permanent belief.

across groups, through the measured shares and the scorecard of Appendix F.

Two features of (8) carry the rest of the section. The first is that the bottom's prior excess does not appear in it: the perceived prior enters the agent's own reading and the recursion's gain by offsetting factors and cancels from the resting mean (Appendix F), so the reading error that inflates the prior survives in the transitory prior $\sigma_{\theta,\text{perc},g}^2$, never in the resting permanent belief. The second is the corollary, that the permanent belief, which is the object driving precautionary saving, moves with legibility alone, so the saving distortion is a legibility story and not a prior one.

The perceived-risk wedge

The object that organises the rest of the section is the gap between the agent's perceived second moments and the true ones, summed across both shocks, which I call the perceived-risk wedge,

$$\Delta\sigma^2 = (\hat{\sigma}_\psi^2 - \sigma_\psi^2) + 2(\sigma_{\theta,\text{perc}}^2 - \sigma_\theta^2), \quad (9)$$

the permanent gap plus twice the transitory gap, in the units of the realised process. Its two pieces are signed differently, and that difference generates the cross-section. The permanent gap $\hat{\sigma}_\psi^2 - \sigma_\psi^2 = \sigma_\psi^2[V_u/\bar{S}_m - 1] \leq 0$ is one-signed, since legibility can only depress perceived permanent risk, while the transitory gap $\sigma_{\theta,\text{perc}}^2 - \sigma_\theta^2 = \omega_g^2 - \rho_g\sigma_\theta^2$, by (6), is two-signed, lifted by reading noise at the bottom and lowered by netting higher up. Because the two gaps pull in opposite directions, perceived total risk can sit on either side of the truth, and across groups it crosses over: the prior excess dominates at the bottom, where the wedge is positive, and legibility dominates higher up, where it turns negative. This is the prior crossing of Section 7, now stated as a property of $\Delta\sigma^2$. The wedge (9) is a monthly object; at the twelve-month horizon the permanent gap is scaled by twelve against the transitory's two, so the bottom's total lands within a percent of the benchmark (Table 12) and the crossing lives in the components, not the total.

A gap in second moments reaches behaviour through the curvature of the value function. Writing the continuation value $V(a, y)$ under the agent's perceived income density $\tilde{f}$ in place of the true f , and noting that the two share a conditional mean and differ only in spread, by $\Delta\sigma^2$, the perceived-minus-true continuation is

$$(\Psi V)(a, y) = \int V(a, y') [\tilde{f} - f] dy' \approx \frac{1}{2} \Delta\sigma^2 \partial_{yy}^2 V(a, y), \quad (10)$$

the first-order term cancelling with the means and the leading survivor being the variance error times the second income-derivative of V , with the $\frac{1}{2}$ a Taylor weight and not a free parameter. Equation (10) is a leading-order characterisation, exact in the linear-quadratic limit and otherwise a second-order approximation whose remainder is third-order in the income innovation; the model of Section 9 confirms its magnitude rather than its functional form. I rest the argument that follows not on it but on the prudence term of the Euler equation, taken up in Section 8.4, to which (10) is the compact complement.

The comparative statics of the next subsection make precise which part of the belief block carries the wedge, and the two subsections that follow trace it through prudence and through the belief's own dynamics.

Comparative statics in legibility

Every object in the belief block moves through the unlabelled residual variance V_u , which legibility lowers ($\partial V_u / \partial \rho_g = -2\sigma_\theta^2$), the total $\bar{S}_m$ staying fixed. Differentiating the permanent belief (8), the gain through its observation noise $R_g = 2(\sigma_\psi^2)^2(V_u/\bar{S}_m)^2$ (Appendix F), and the attribution (7) through V_u gives the signs collected in Table 3, where $g_p \equiv \hat{\sigma}_\psi^2 - \sigma_\psi^2$ is the saving-relevant gap.

Object	$\partial/\partial\rho_g$	$\partial/\partial\omega_g^2$	Reaches behaviour?
Permanent belief $\hat{\sigma}_\psi^2$	—	0	yes, the saving margin
Transitory prior $\sigma_{\theta,\text{perc},g}^2$	— via netting	+ one for one	second-order, buffered within the period
Gain μ_g (persistence $1 - \mu_g$)	+	0	dynamics only
Attribution $\lambda_g(\rho_g)$	—	0	news response only
Saving-relevant gap $g_p = \hat{\sigma}_\psi^2 - \sigma_\psi^2$	—	0	yes

Table 3: Comparative statics of the beliefs and the estimated rule in legibility ρ_g and the identified reading noise ω_g^2 of (6). Legibility moves every object; the bottom’s prior excess enters only the transitory prior. The zeros in the ω^2 column hold for the closed forms; in the recursion’s own units the gain λ^{rule} falls with ω^2 by exactly the factor that offsets the inflated reading, the product invariant by construction (Appendix F).

The table makes two points. First, legibility plays two roles: it sets the dynamics, raising the gain and lowering the attribution ($\partial\mu_g/\partial\rho_g > 0$, $\partial\lambda_g/\partial\rho_g < 0$), and it is at the same time the only source of under-perceived permanent risk ($\partial g_p/\partial\rho_g < 0$), so a single primitive generates the persistence of beliefs, the response to news, and the saving distortion together. Second, the bottom’s prior excess reaches the saving margin only at second order, since it is absent from the filter’s signal-to-noise ratio, the closed-form attribution, and the resting permanent belief, the recursion’s own gain λ^{rule} falling with ω^2 by the exactly offsetting factor (Appendix F); the decomposition of Section 10 confirms it, putting the prior’s contribution to the wedge at nil. The saving-relevant gap is therefore a ρ_g -statistic, identical across two groups that read their pay equally legibly, however differently their priors are pinned, which is why the behavioural wedge widens with ρ_g along the income gradient and not with the prior.

Prudence and the incidence of the wedge

A belief about a variance moves saving through a different part of preferences than a belief about the level would. Writing $\widetilde{\mathbb{E}}$ and $\widetilde{\text{Var}}$ for the expectation and variance under the agent’s belief $\hat{\sigma}_\psi^2$, and expanding expected marginal utility around mean next-period consumption, $\widetilde{\mathbb{E}}[u'(c')] \approx u'(\widetilde{\mathbb{E}}c') + \frac{1}{2}u'''(\widetilde{\mathbb{E}}c')\widetilde{\text{Var}}(c')$, a belief about the mean would move the first term through u'' , the curvature that fixes the elasticity of intertemporal substitution, while a belief about the variance touches only the second, through u''' . A perceived-variance error $\hat{\sigma}_\psi^2 - \sigma_\psi^2$ therefore acts on saving through prudence, just as a level error would act through the elasticity of intertemporal substitution. Carrying the consumption policy through,

$$\widetilde{\mathbb{E}}[u'(c')] - \mathbb{E}[u'(c')] \approx \frac{1}{2}(\hat{\sigma}_\psi^2 - \sigma_\psi^2)\mathbb{E}[u'''(c')c_y^2 + u''(c')c_{yy}], \quad (11)$$

with c_y and c_{yy} the first two income-derivatives of the consumption policy. Under the CRRA used here ($\gamma = 2$, so $u''' = \gamma(\gamma + 1)c^{-\gamma-2} > 0$) the leading term is positive, so an agent who perceives

less permanent risk than she faces under-saves for precaution, opening the wealth and marginal-propensity-to-consume wedge that Section 10 quantifies (Tables 12 and 13). The operative gap is the permanent one: a permanent surprise accumulates in the random walk and shifts the whole future consumption path, whereas a transitory belief error is buffered within the period and barely reaches c_y .

The incidence of this channel is gated by liquidity. Because perceived risk acts only through precautionary saving, the saving response to the permanent-variance belief, $\partial a' / \partial \hat{\sigma}_\psi^2$, is zero at the borrowing constraint $a' = \underline{a}$ and strictly positive only where the household holds slack, the opposite of a level belief error, which reaches a constrained, high-MPC household one for one. In the hard-constraint limit the two incidences are disjoint: a variance error spares the constrained and bites the unconstrained, a level error the reverse. In the model's cross-section, however, the ordering of the wedge is set by the size of the permanent-belief gap rather than by the constraint. The gap is a legibility statistic, small at the bottom, which labels little ($\rho_g = 0.12$, so the discount in (8) is 0.96), and largest at the top, and Section 10 finds the wealth wedge per unit of belief gap nearly flat across groups, so liquidity reinforces the incidence at the margin without creating it. This is why the bottom can hold the most distorted transitory prior of any group and yet bear the smallest behavioural wedge, the cross-section Table 13 quantifies: its distortion sits in the prior, which is buffered within the period, while its permanent belief is essentially unbiased.

Level shocks versus risk shocks

The belief carries its own dynamics, and they decide whether a shock leaves a lasting wedge. The permanent belief is a state driven by the squared surprise x_t , $\hat{\sigma}_{\psi,t}^2 = (1 - \mu)\hat{\sigma}_{\psi,t-1}^2 + \mu\lambda(\rho)x_t$, so a single surprise propagates as $\partial \hat{\sigma}_{\psi,t+h}^2 / \partial x_t = \mu\lambda(\rho)(1 - \mu)^h$, a jump of $\mu\lambda(\rho)$ that decays geometrically at rate $1 - \mu$, and the saving wedge inherits this kernel. A level shock, the kind a recession delivers, is a single squared surprise, so it traces the transient kernel and reverts, within a quarter at the calibrated gains, leaving no lasting wedge. A risk shock, a persistent rise Δ in the true permanent variance, enlarges the squared surprises for as long as it lasts; under the constant gain the agent books only the share $\lambda(\rho)$ of that increase as permanent, so her perceived permanent variance rises by $\lambda(\rho)\Delta$ and falls permanently short of the truth by the complement $(1 - \lambda(\rho))\Delta$. The shortfall is largest at the top, which attributes the least of any surprise to permanent risk.⁵ The behavioural response again concentrates at the top, and for the steady state's reason: the top's shortfall is the largest because it attributes the least, while the bottom, attributing nearly its full rational share, hardly under-perceives the rise at all. Section 9 embeds the measured beliefs in the quantitative model, and Section 10 takes both shocks to it.

9 A heterogeneous-agent model

To trace what those perceptions imply for behaviour, I compare each group under its estimated beliefs with an otherwise identical full-information benchmark in a standard heterogeneous-agent

⁵Stated in the units of the full squared change, the form of (2) with the attribution (7). Written on the agent's own netted reading (5), the recursion's gain is the rescaled λ^{rule} (Appendix F); the two forms share the resting mean by construction but not the point response, and reading noise, silent in the steady state, damps the bottom's response in the implemented form. The quantified impulse responses are computed in the implemented form and will state the passthrough in its units.

quantitative model. The gap between the two regimes, in precautionary wealth and the marginal propensity to consume, is the belief wedge.

Setup

I embed the estimated belief structure in a life-cycle buffer-stock economy following Wang (2023), solved in partial equilibrium, with a one-asset general-equilibrium closure in Section 10. The discount factor is a single value common across agents, calibrated to average liquid wealth (the calibration below takes this up), so the cross-section of wealth and of marginal propensities to consume is an outcome of the income and employment risk rather than of preference heterogeneity. The model is monthly and overlapping-generations: a worker enters at age 25, works for forty years, retires at 65 onto a pension replacing sixty percent of her permanent income, and dies with certainty at 85, facing a small constant mortality risk before then; each death is matched by a new entrant drawn across the income distribution, so the cross-section is stationary through the finiteness of life rather than through the unit-root death-and-replacement device an infinite-horizon permanent random walk requires. Preferences are time-separable with constant relative risk aversion $\gamma = 2$ and a homothetic warm-glow bequest in terminal assets.

Income follows the permanent-transitory process of Section 4 scaled by a deterministic age-wage profile, $y_{i,t} = \exp(p_{i,t}) G_\tau \xi_{i,t}$, with $p_{i,t} = p_{i,t-1} + \psi_{i,t}$ the permanent random walk, G_τ a standard hump-shaped age profile, and $\xi_{i,t}$ the transitory state, the variances taken from Table 2. When employed the worker draws $\xi = \exp(\theta_{i,t})$; when unemployed she receives a benefit replacing half her permanent income; in retirement she receives the pension. The permanent innovations $\psi_{i,t}$ are the same ones she learns about in Section 5: accumulated over a career they carry her across the income distribution, so a run of positive permanent raises lifts a worker out of the bottom group without any change of job, the deterministic profile G_τ adding a predictable component to the same drift; a current-income tercile therefore mixes types, a distinction the results return to. This is mobility in the sense the same-job perceived-risk question allows, the gradual drift of earning power rather than job-switching, consistent with the within-job conditioning of the elicitation. The discount factor is common, while the measured labelled share, the update gain, and the perceived transitory prior are functions of current income, $\rho(y)$, $\mu(y)$ and $\sigma_{\theta, \text{perc}}^2(y)$, interpolated through the tercile anchors of Sections 6 and 7, and a worker slides along them as her income drifts. Following Carroll (2006) I normalise by permanent income: the state is cash-on-hand per unit of permanent income, the per-period draw is the transitory $\exp(\theta)$, and permanent shocks enter through the growth factor $G_\tau \exp(\psi)$. The problem is scale-invariant, so I report each group's mean monthly SCE income (\$2,415 / \$6,014 / \$13,604) only to fix magnitudes.⁶

Two regimes. I solve the model twice. Under the *belief* regime the agent perceives wage risk through the estimated rule and her belief evolves; under *FIRE* she knows the true variances. Everything else is shared, so any gap is the belief wedge. FIRE here means correct wage-risk beliefs; job-loss risk is common to both regimes and nets out.

Unemployment. The SCE measures *same-job* risk, so the budget adds a two-state employment block whose rates I take from the SCE itself, parallel to the wage-risk measurement and following

⁶Wage variances are stated in $(\%)^2$; in the income process they enter in fractional units (divided by 10^4) before exponentiating, and the perceived-variance state is carried back in $(\%)^2$.

Wang (2023): each respondent’s perceived probability of losing her job within twelve months and of finding one within three months were she to lose it, survey-weighted by income tercile over employed respondents and converted to monthly Poisson rates.⁷ This gives monthly separation 0.015/0.011/0.011 and job-finding 0.23/0.27/0.29 from bottom to top, a steady-state unemployment rate of 6.2/4.0/3.6 percent, higher and more persistent at the bottom, with replacement income half of permanent. The rates sit a little below the realised aggregate (a Shimer (2005) benchmark near seven percent), the known gap between perceived and realised job-loss risk; what the model uses is the perceived gradient, common to both belief regimes and perceived correctly, so it shapes the wealth distribution and the hand-to-mouth share, in the manner Wang (2023) emphasises; it is not a source of the wedge, though it shapes the constraint incidence through which the wedge acts. Beliefs update only while employed and freeze during unemployment.

The household problem

$$V_\tau(m, p, s, \hat{\sigma}_\psi^2) = \max_{0 \leq c \leq m} \left\{ u(c) + \beta (1-D) \tilde{\mathbb{E}}[V_{\tau+1}(m', p', s', \hat{\sigma}_{\psi'}^2)] \right\}, \quad m' = \frac{R}{\Gamma'} (m-c) + \xi', \quad \Gamma' = G_\tau \exp(\psi'),$$

with normalised cash-on-hand m , age τ , income node p , employment status s , monthly mortality D , and $\tilde{\mathbb{E}}$ the expectation under the agent’s *perceived* process; the terminal age carries the warm-glow bequest in place of a continuation value. The budget uses *realised* income while expectations are taken under the perceived process, so beliefs act only through the saving motive, and a higher $\hat{\sigma}_\psi^2$ raises perceived risk and hence precautionary saving through prudence ($u''' > 0$, Section 8). The agent observes realised income each period but cannot decompose a given month’s change into permanent and transitory parts, the signal-extraction problem of Section 7; for tractability the budget is written in permanent-income-normalised form using her running estimate of permanent income, so the normalisation is an approximation, not a claim that she observes the permanent component exactly. Beliefs then distort the perceived *variance* of future shocks, not the level of current income. The belief follows the constant-gain rule of Section 5, $\hat{\sigma}_{\psi,t}^2 = (1 - \mu(y))\hat{\sigma}_{\psi,t-1}^2 + \mu(y)\lambda(y)(\Delta\tilde{y}_t)^2$, on the agent’s own netted reading $\tilde{y}$ of (5), where the gain on that reading is not free but the closed-form attribution (7) at the measured labelled share $\rho(y)$, rescaled to the reading’s variance (λ^{rule} , Appendix F, Table 11), while the update speed $\mu(y)$ and the perceived transitory prior are carried as the estimated moments themselves (Section 7); it settles around $\lambda(y) \mathbb{E}[(\Delta\tilde{y})^2] = \sigma_\psi^2 V_u / \bar{S}_m$, the labelling discount holding it below the realised permanent variance wherever pay is partly labelled, mildly at the bottom and most at the top. The steady-state comparison below uses the resting value of this rule, its long-run mean, which adds no state dimension; the dynamic state is carried explicitly for the risk-shock response.

Stationary equilibrium. For each regime a stationary equilibrium is a family of age-indexed value functions, policies $c_\tau(m, p, s, \hat{\sigma}_\psi^2)$, and an invariant distribution Φ such that the policies solve the household problem at the fixed r and Φ is invariant under them, the income process including the permanent drift across groups, the life-cycle demographics of entry, retirement, mortality and

⁷For each employed respondent I take the SCE’s reported percent chance of losing the main job over the next twelve months and, conditional on losing it, of finding a new one within three months, average each within income tercile with survey weights, and convert a horizon- H probability P to a monthly hazard $-\log(1-P)/H$ (with $H = 12$ for separation and $H = 3$ for finding), the conversion Wang (2023) applies to the same SCE questions. The tercile-mean twelve-month job-loss probabilities are 16.9/12.7/12.2 percent and the three-month finding probabilities 50.2/56.0/57.6 percent.

replacement, the employment transitions, and the belief law of motion (degenerate at the truth under FIRE). I read moments of Φ off a simulated panel of $N = 30,000$ households over 840 months (480 burn-in), with the belief and FIRE branches driven by a single common shock path, so each worker sits in the same income tercile under both regimes and the belief-minus-FIRE gap is a within-worker difference free of sampling noise.

Calibration

The calibration sets a single discount factor, common across agents and calibrated to average liquid wealth, and lets the belief block vary with income through the labelled share ρ and the carried estimates of Section 6; the share is the block's one pinned parameter, set jointly by its measurement band and the belief moments below, and nothing else in the block is fitted. The model is block-recursive: beliefs evolve on the realised income process and never on assets or consumption, so the belief block is measured and estimated independently of preferences. Table 4 collects the calibration.

Parameter	Symbol	Value (Bot / Mid / Top)	Source or target
<i>A. Preferences and assets (common)</i>			
Risk aversion	γ	2	Assigned (standard)
Annual interest rate	r	2%	Assigned (standard)
Borrowing limit	$\underline{a}$	0	Assigned (standard)
Discount factor	β	0.879	Calibrated to mean liquid wealth (≈ 1.8 mo)
<i>B. Employment block (heterogeneous by income; common across regimes)</i>			
Separation rate	s_g	0.015/0.011/0.011/mo	SCE perceived (Q13new), monthly Poisson
Job-finding rate	f_g	0.23/0.27/0.29/mo	SCE perceived (Q22new), monthly Poisson
Replacement rate	b	0.5	Standard convention
Pension replacement	ς	0.6	Standard convention
<i>C. Realised income process by group, (%)²</i>			
Permanent variance	σ_ψ^2	11.9/11.6/11.2	SIPP (Table 2)
Transitory variance	σ_θ^2	2.5/4.4/5.8	SIPP (Table 2)
Mean monthly income	—	\$2,415/6,014/13,604	SCE (scale anchor only)
<i>D. Beliefs by group: the labelled share pinned within its measured band, the rest implied or carried</i>			
Labelled share	ρ_g	0.119/0.197/0.369	Primitive: SIPP band, reconciled ($J = 1.57$, $p = 0.46$)
Update gain	μ_g	0.629/0.910/0.950	Estimated GMM moment, carried (Section 6)
Attribution	λ_g	0.677/0.519/0.398	Implied from ρ_g , eq. (7) (App C)
Transitory prior	$\sigma_{\theta, \text{perc}, g}^2$	9.65/3.55/4.12	GMM estimate, carried; at its netting floor at Mid

Table 4: Model calibration. Panel A: a single discount factor common across agents, calibrated so the FIRE economy matches average liquid wealth (≈ 1.8 months); B: the employment block, perceived job-separation and job-finding from the SCE (Section 9), heterogeneous by income but common to both regimes; C: the realised SIPP process; D: the belief regime, where the labelled share ρ_g is pinned jointly, the measured SIPP band supplying the level and the attribution ratios and prior floors selecting within it ($J = 1.57$, $p = 0.46$); the attribution follows from it in closed form, the gain is carried as the estimated GMM moment, and the prior is carried as the estimate where it clears its netting floor and at the floor where it does not (the middle); under FIRE beliefs are held at the truth. The measured share carries a specification band, from an imputation-dropped lower edge to a detrended upper edge (Appendix F), which Section 10 carries through to the wedge.

Preferences, a single discount factor to average liquid wealth. Preferences are CRRA ($\gamma = 2$, $r = 2$ percent annual). I calibrate a single discount factor, common across agents, so that the FIRE economy matches the average liquid wealth in the data, about 1.8 months of permanent income across the three groups, which gives an annual β of 0.879. This is the standard incomplete-markets discipline: it pins the *level* of wealth and so of the marginal propensity to consume, while leaving the cross-section of both an outcome of the income and employment risk. It is a middle course between the two alternatives. Fixing the discount factor at a consensus value, as Wang (2023) does, leaves the level an outcome and the economy over-accumulates, with liquid wealth running to tens of months and marginal propensities far below the data; calibrating an income-varying $\beta(y)$ to each group's wealth target instead builds the cross-sectional gradient into preferences. A single calibrated discount factor avoids both, hitting the average and letting the cross-section emerge as Section 10 reports it.⁸ Calibrating on the FIRE regime is a normalisation, not a substantive choice: under the measured beliefs average wealth falls short of the FIRE target by a few percent, so recalibrating β on the belief regime would move it only marginally and leave the wedge essentially unchanged. The discount factor itself sits below the 0.95–0.96 of the life-cycle literature (Gourinchas and Parker, 2002), the price of holding a one-asset economy to 1.8 months of *liquid* wealth rather than to total net worth, while the twelve-month marginal propensities of 0.48–0.56 and hand-to-mouth shares of 28–41 percent lie inside the empirical ranges of Kaplan et al. (2014). I solve by the endogenous grid method (Carroll, 2006) and simulate each regime on a common shock path.

Beliefs, pinned and carried. The labelled share is pinned by reconciliation: the SIPP measurement band of Section 7 supplies the level, and the belief moments select within it. The share triple minimises the standard-error-weighted distance to three sets of moments, the measured shares themselves with the specification band as their uncertainty, the estimated attribution ratios, and the admissibility of the carried priors against their netting floors $(1 - \rho_g)\sigma_\theta^2$, which the mechanism does not let a prior undercut. The reconciled shares are 0.119/0.197/0.369, none more than a quarter of the band's standard error from its measured centre. The admissibility constraint moves one carried moment: the middle prior's estimate, 2.96, sits below its floor at every share in the band, so the fit carries that prior at the floor, 3.55, six tenths of a standard error from the estimate, while the bottom and top estimates clear their floors and are carried as estimated, 9.65 and 4.12. The fit is overidentified, and the test reads $J = 1.57$ ($p = 0.46$): the pay records, the belief responses, and the prior floors are mutually consistent, which is the mechanism's test. The attribution the household block reads is the closed form (7) at the reconciled shares, 0.677/0.519/0.398; the gain is carried directly as the estimated GMM moment of Section 6, and the prior enters with its mild non-monotonicity intact, since the floors predict it. The perceived permanent variance that drives saving then follows from the identity (8), a discount $V_u/\bar{S}_m$ of 0.96/0.91/0.81 on the realised permanent variance. Table 5 collects the pinned, implied, and carried objects side by side. The estimated attribution level, 0.079/0.046/0.026, is not comparable to the structural one, the cell-mean regressor attenuating it by an order of magnitude; the comparable moments are the cross-group ratios, which the reconciled shares predict to within one standard error, at about half the point gradient, the residual half being under-reaction beyond labelling or sampling noise (the

⁸Targets from the SCE Household Finance module, the liquid-wealth concept of Kaplan et al. (2014): median liquid wealth about 0.5/2.0/3.0 months across the three groups, averaging near 1.8.

scorecard of Appendix F).

Group	Pinned	Implied			Carried	
	ρ_g	$V_u/\bar{S}_m$	$\hat{\sigma}_\psi^2$	λ_g	μ_g	$\sigma_{\theta,\text{perc},g}^2$
Bot	0.119	0.965	11.47	0.677	0.629	9.65
Mid	0.197	0.915	10.61	0.519	0.910	3.55
Top	0.369	0.812	9.11	0.398	0.950	4.12

Table 5: The belief block: pinned, implied, and carried. The labelled share ρ_g is pinned within its SIPP measurement band by the belief moments (this section); the discount $V_u/\bar{S}_m$, the resting permanent belief $\hat{\sigma}_\psi^2 = \sigma_\psi^2 V_u/\bar{S}_m$ of (8), and the attribution λ_g of (7) follow from it in closed form; the gain is the GMM estimate of Section 6 carried directly, and the prior the estimate carried where it clears its netting floor and at the floor where it does not (Mid); the bottom’s carried prior is its floor plus the identified reading noise $\hat{\omega}^2 = 7.43$ of (6). Variances in $(\%)^2$.

Measurement and its band. The measured shares rise with income, as the mechanism and the performance-pay literature predict (Lemieux et al., 2009). The uncertainty that matters for them is specification rather than sampling, given the 2.25 million stayer months behind the point estimates: the main specification demeans each pay component within a job spell and takes the labelled share as the share of within-spell transitory variance that bonuses and commissions account for, and the alternatives bracket it, from a lower edge that drops months with imputed component amounts (0.042/0.085/0.314) to an upper edge that detrends rather than demeans each spell (0.176/0.256/0.488), the gradient surviving in every specification (Appendix F); Section 10 carries this band through to the wedge. The gradient has a direct counterpart in the same SIPP sample: the share of workers receiving a bonus rises from 2.8 to 4.8 to 9.7 percent across the groups in 2018 (2.8/7.1/12.1 in 2021), receipt of any labelled component from 7.9 to 16.5 percent, while tips, the unlabelled kind, fall from 2.4 to 0.9. Receipt shares speak to the direction of the delivery gradient, not to the level of ρ , and they place the middle below the top on every labelled component, as the measured shares do.

What the measured beliefs imply. Figure 6, reported numerically with its decomposition in Table 12 (Appendix G), sets the stationary perceived variance against FIRE, in the same SIPP units, so the σ_ψ^2 inflation that blocks an absolute comparison with the SCE (Section 4) cancels. What the model carries from the SCE is therefore its cross-group ratios, its dynamics, and the prior gradient, never the absolute level, which the two surveys do not share. Perceived twelve-month risk falls with income, from 153 through 139 to 124, against a FIRE benchmark flat near 148: the bottom perceives about three percent more total risk than it faces, its inflated transitory prior outweighing a tercile-mean permanent discount of about four percent, while the middle and the top perceive six and sixteen percent less, the labelling discount deepening with income faster than the prior falls. The crossing sits in the components, as Section 8 derives: the permanent-variance belief that drives saving is discounted least at the bottom and most at the top, while the prior excess lives at the bottom alone. The current-income bottom tercile mixes in higher-income workers on a weak month, so a bottom-income *type*, which labels little ($\rho = 0.12$), carries a nearly unbiased permanent belief and little saving-relevant distortion. The saving wedge the bottom *tercile* still shows in Table 13 is partly composition: just over half the tercile is retirees, whose beliefs no longer

update, and part of the remainder is higher-income workers passing through on a weak month; the workers-only panel of Table 13 restricts to working households and preserves the same shape.

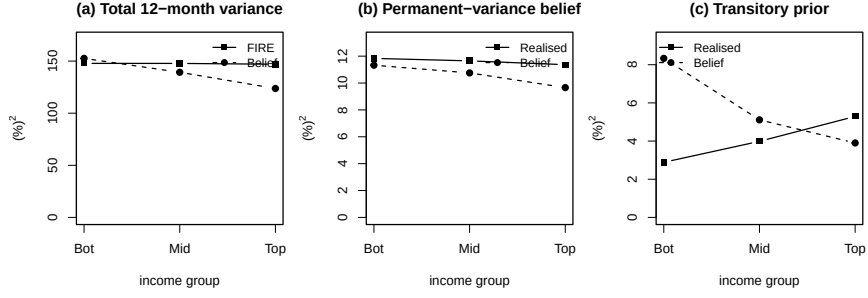


Figure 6: Total perceived variance under the measured beliefs at the 12-month horizon against the FIRE benchmark, by income group, in $(\%)^2$; the total is $12\hat{\sigma}_\psi^2 + 2\sigma_{\theta,\text{perc}}^2$. The permanent-belief and transitory-prior decomposition behind it is in Table 12.

10 Partial-equilibrium results

Steady state

Figure 7 compares the belief regime with an otherwise identical FIRE benchmark; the underlying numbers are in Table 13 (Appendix G). With the discount factor calibrated to average liquid wealth, the FIRE economy holds 2.3/1.7/1.5 months of permanent income, with twelve-month MPCs of 0.56/0.50/0.48 and hand-to-mouth shares of 41/29/28 percent. The propensities and the shares are in the data's range; mean wealth in months falls mildly with income where the median in the data rises, a miss that survives the workers-only split of Table 13 and that the single common discount factor accepts as the price of leaving the cross-section an outcome. The object of interest is the belief wedge, computed with this calibration held fixed.

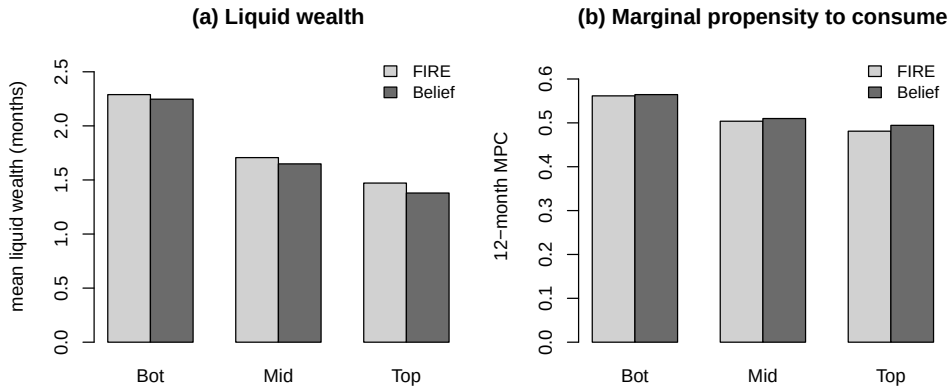


Figure 7: Mean liquid wealth (months of permanent income) and the 12-month MPC by income group, FIRE versus the belief regime (Table 13).

The wedge in wealth rises monotonically with income, the belief economy holding 1.9, 3.4, and 6.3 percent less than FIRE from bottom to top under the reconciled shares, and 2.6, 4.5, and 8.1 percent at the detrended upper edge of the measurement band (Appendix F), while the wedge in the marginal propensity to consume concentrates at the top, +0.3, +0.6, and +1.3 points. Both

profiles track the same object, the permanent-belief gap, smallest at the bottom and largest at the top: per unit of that gap the wealth wedge is broadly flat across groups, at three to four percent of wealth per $(\%)^2$, so the cross-section of the wedge is essentially the cross-section of the gap, and liquidity reinforces the incidence at the margin without creating it. A decomposition makes the same point exactly. Solving the model with the permanent discount alone, the transitory prior held at the truth, reproduces the full wedge on both margins, within two percent of it in wealth and exactly in the marginal propensity; the complementary regime, the carried prior with the permanent belief held at the truth, moves neither wealth nor the marginal propensity measurably. The prior is behaviourally inert in this economy; the discount is the wedge.

The bottom is barely moved on either margin, +0.3 points on the MPC and 1.9 percent on wealth, exactly as Section 8 anticipates: a bottom-income type labels little ($\rho = 0.12$), so its permanent belief is discounted by under four percent and carries little saving-relevant distortion. Composition does part of the rest. Just over half of the bottom current-income tercile is retirees, against roughly a quarter and a tenth above, and among its workers the bottom is in fact the least constrained of the three groups, with hand-to-mouth shares of 14/21/23 percent from bottom to top, so the small wedge is a fact about beliefs and composition, not about a binding constraint; restricting to workers and re-terciling among them, the wedge is $-2.2/ -4.1/ -7.0$ percent, the same shape mildly sharpened. The middle is moderately distorted on both margins (-3.4 percent on wealth, +0.6 points on the MPC): unconstrained and labelling more of its transitory pay, it under-saves. The top responds most on both, holding 6.3 percent less wealth and consuming over a point more of a windfall over the year, the group that labels its transitory pay away most completely, under-perceives the permanent risk that drives saving most, and acts on it. The labelled belief lies below the truth at all times, so the wedge is a steady-state object, not merely a response to shocks.

An upper bound. The measured calibration reads the belief estimates through the measured delivery of pay; a variant instead takes the estimated attribution responses at face value, anchoring the bottom at its full-information share and letting the estimated attribution ratios set the middle and the top, $\lambda = 0.702/0.409/0.231$, so the resting permanent belief falls to 11.89/8.36/5.29, discounts of 1.00/0.72/0.47 against the reconciled 0.96/0.91/0.81 (Appendix F). Its wedge is 3.1, 8.8, and 16.5 percent of wealth and +0.4, +1.6, and +3.8 points on the marginal propensity, the same shape at roughly three times the size. The variant shows what the estimated belief responses imply when delivery is asked to carry the whole of their gradient; the gap between it and the measured calibration is the part of the wedge that rides on netting beyond the measured pay components, or on sampling noise in attribution ratios whose standard errors admit either reading.

General equilibrium

Closing the economy lets the saving wedge move the interest rate. Output is produced with a constant-returns technology, $Y = ZK^\alpha N^{1-\alpha}$ with capital share $\alpha = 0.36$ and monthly depreciation from an annual 2.5 percent, fund the unemployment benefit and the pension with flat balanced-budget taxes on labour income, and pin the productivity normaliser Z once from the FIRE economy at a target rate of three percent, holding it fixed across regimes so the FIRE-versus-belief rate gap is the general-equilibrium wedge. At the calibrated discount factor the FIRE economy clears at three percent by construction; the belief economy, under-perceiving permanent risk, supplies

about three percent less aggregate capital and clears at a rate 0.12 percentage points higher, 3.1 against 3.0 percent, with the wealth Gini 0.003 higher (Table 14, Appendix G). This is the choice channel of Wang (2023) in general equilibrium: lower perceived risk lowers precautionary supply, raises the equilibrium return, and concentrates wealth, though under the measured shares the aggregate movement is modest, the wedge living in the cross-section more than in the aggregates. The magnitude is the one-asset response, in which a single liquid asset carries both the buffer and the return; a credible quantification of how the wedge moves marginal propensities and the transmission of aggregate shocks needs a two-asset closure separating liquid from illiquid wealth.

Dynamics

The belief is a state, so the wedge carries the dynamics derived in Section 8: a level shock of the recession kind traces the transient kernel and reverts within a quarter, leaving the chronic wedge untouched, while a persistent risk shock leaves a lasting shortfall of $(1 - \lambda(\rho))\Delta$, largest at the top, where the response again concentrates (in the full-change form of Section 8.5; see the footnote there). The dynamic-belief solution carries the belief as a fourth state and reproduces the steady-state wedge as its long-run mean; the quantified impulse responses to the two shocks are in progress.

11 Conclusion

The paper opened by asking whether the risk heterogeneous-agent models assign their agents is the risk those agents perceive, and whether any gap between the two falls evenly across households or pulls them apart. It is not, and it pulls them apart. Conditional on staying in the same job, workers at the bottom of the income distribution report about 1.8 times the wage uncertainty of workers at the top, hold that belief more persistently, and revise it more strongly after a realised shock, against a realised process that shows no counterpart to any of the three. A single constant-gain rule organises the three gradients within person; the persistence and prior gradients are statistically significant, the attribution gradient monotone in point estimates.

The mechanism the paper attaches to the rule is measured, not conjectured: the legibility of pay, the share of transitory pay that arrives labelled and is netted out before the update, is measured directly from the itemised pay components of the same SIPP panel, rising from about an eighth of transitory pay variance at the bottom to over a third at the top, and the estimated belief moments then become predictions it is scored against. The measured netting floors place the middle and top priors within a standard error of where they sit, isolate the bottom’s reading-noise residual at four standard errors from zero, and predict the attribution ratios within a standard error at about half the point gradient, a single primitive generating the persistence of beliefs, their response to news, and the saving distortion together, with nothing in the belief block fitted. Embedded in a life-cycle economy under one calibrated discount factor, the measured beliefs open a precautionary wedge that rises with income and concentrates at the top, the group whose perceived permanent risk is discounted most: 2, 3, and 6 percent of liquid wealth and up to 1.3 points of the twelve-month MPC in partial equilibrium, reaching 8 percent and 1.7 points at the upper edge of the measurement band and 16 percent and 3.8 points where the estimated belief responses are taken

at face value, and in one-asset general equilibrium a capital stock 3 percent lower and an interest rate 0.12 points higher than under full information, the wedge living in the cross-section more than in the aggregates.

The natural test of the mechanism is out of sample, in variation in how pay is delivered rather than in how much it varies: performance-pay shares across occupations and reforms to payroll formats move legibility while holding risk fixed, and the rule says whose beliefs they should move. The test is sharper now that the primitive is observable: a payroll reform shifts a labelled share that can be measured before and after in the same records, so the comparison is between two measurements and a predicted belief response, not between a fitted parameter and a hoped-for proxy. The quantified impulse responses from the dynamic belief state, and a two-asset closure in which the wedge's effect on marginal propensities can be credibly priced, remain for future work.

References

- Blundell, R. and Bond, S. (1998). Initial conditions and moment restrictions in dynamic panel data models. *Journal of Econometrics*, 87(1):115–143.
- Bound, J. and Krueger, A. B. (1991). The extent of measurement error in longitudinal earnings data: Do two wrongs make a right? *Journal of Labor Economics*, 9(1):1–24.
- Broer, T., Kohlhas, A., Mitman, K., and Schlafmann, K. (2022). Expectation and wealth heterogeneity in the macroeconomy. Working paper.
- Caplin, A., Gregory, V., Lee, E., Leth-Petersen, S., and Sæverud, J. (2023). Subjective earnings risk. NBER Working Paper 31019, National Bureau of Economic Research.
- Carroll, C. D. (2006). The method of endogenous gridpoints for solving dynamic stochastic optimization problems. *Economics Letters*, 91(3):312–320.
- Gourinchas, P.-O. and Parker, J. A. (2002). Consumption over the life cycle. *Econometrica*, 70(1):47–89.
- Grigsby, J., Hurst, E., and Yildirmaz, A. (2021). Aggregate nominal wage adjustments: New evidence from administrative payroll data. *American Economic Review*, 111(2):428–471.
- Guerreiro, J. (2023). Belief disagreement and business cycles. Job Market Paper, Northwestern University.
- Guvenen, F., Karahan, F., Ozkan, S., and Song, J. (2021). What do data on millions of US workers reveal about lifecycle earnings dynamics? *Quarterly Journal of Economics*, 136(4):2349–2391.
- Halvorsen, E., Holter, H. A., Ozkan, S., and Storesletten, K. (2024). Dissecting idiosyncratic earnings risk. *Review of Economic Studies*.
- Hardy, B. and Ziliak, J. P. (2014). Decomposing trends in income volatility: The “Wild Ride” at the top and bottom. *Economic Inquiry*, 52(1):459–476.
- Hoynes, H., Miller, D. L., and Schaller, J. (2012). Who suffers during recessions? *Journal of Economic Perspectives*, 26(3):27–48.
- Kaplan, G., Violante, G. L., and Weidner, J. (2014). The wealthy hand-to-mouth. *Brookings Papers on Economic Activity*, pages 77–138.
- Kaufmann, K. and Pistaferri, L. (2009). Disentangling insurance and information in intertemporal consumption choices. *American Economic Review: Papers & Proceedings*, 99(2):387–392.
- Kim, G. and Binder, C. (2023). Learning-through-survey in inflation expectations. *American Economic Journal: Macroeconomics*, 15(2):254–278.
- Kuchler, T. and Zafar, B. (2019). Personal experiences and expectations about aggregate outcomes. *Journal of Finance*, 74(5):2491–2542.
- Lemieux, T., MacLeod, W. B., and Parent, D. (2009). Performance pay and wage inequality. *Quarterly Journal of Economics*, 124(1):1–49.
- Ma, Y., Ropele, T., Sraer, D., and Thesmar, D. (2020). A quantitative analysis of distortions in

- managerial forecasts. NBER Working Paper 26830, National Bureau of Economic Research.
- Meghir, C. and Pistaferri, L. (2004). Income variance dynamics and heterogeneity. *Econometrica*, 72(1):1–32.
- Morduch, J. and Schneider, R. (2017). *The Financial Diaries: How American Families Cope in a World of Uncertainty*. Princeton University Press.
- Pischke, J.-S. (1995). Individual income, incomplete information, and aggregate consumption. *Econometrica*, 63(4):805–840.
- Rozsypal, F. and Schlafmann, K. (2023). Overpersistence bias in individual income expectations and its aggregate implications. *American Economic Journal: Macroeconomics*, 15(4):331–371.
- Shimer, R. (2005). The cyclical behavior of equilibrium unemployment and vacancies. *American Economic Review*, 95(1):25–49.
- Wang, T. (2023). Perceived versus calibrated income risks in heterogeneous-agent consumption models. Bank of Canada Staff Working Paper 23-59.

A From EWMA to observable AR(1)

Solving the level identity (3) for the belief gives $\hat{\sigma}_{\psi,t}^2 = (\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2)/12$. Substituting both belief terms into (2) and multiplying by 12,

$$\bar{\sigma}_{\text{perc},t}^2 - 2\sigma_{\theta,\text{perc}}^2 = (1 - \mu)[\bar{\sigma}_{\text{perc},t-1}^2 - 2\sigma_{\theta,\text{perc}}^2] + 12\mu\lambda(\Delta e_{t-1})^2.$$

Collecting the intercept ($2\mu\sigma_{\theta,\text{perc}}^2$), adding group subscripts, a person fixed effect α_i , and a shock $u_{i,t}$ gives (4).

B 12-month FIRE variance

Rolling $e_t = p_t + \theta_t$ forward H months, the permanent piece accumulates one fresh innovation per month, $\text{Var}(p_{t+H} - p_t) = H\sigma_{\psi}^2$, while the iid transitory piece contributes only its endpoints, $\text{Var}(\theta_{t+H} - \theta_t) = 2\sigma_{\theta}^2$. The shocks are independent, so $\text{Var}(e_{t+H} - e_t) = H\sigma_{\psi}^2 + 2\sigma_{\theta}^2$, which at $H = 12$ is (1).

Group	FIRE benchmark ($12\sigma_{\psi}^2 + 2\sigma_{\theta}^2$)	SCE observed $\bar{\sigma}_{\text{perc}}^2$	ratio
Bot	148.1	20.6	7.2×
Mid	148.2	11.3	13.1×
Top	146.4	11.4	12.8×

Table 6: FIRE-implied 12-month variance versus SCE-reported perceived variance, in (%)².

C Closed form for λ_g under partial labelling

Split the transitory shock $\theta_t = \theta_t^{\text{lab}} + \theta_t^{\text{unlab}}$ into independent pieces with variances $\rho_g\sigma_{\theta}^2$ and $(1 - \rho_g)\sigma_{\theta}^2$. Then $\Delta e_t = u_t + l_t$, with unlabelled residual $u_t = \psi_t + \theta_t^{\text{unlab}} - \theta_{t-1}^{\text{unlab}}$ and labelled component $l_t = \theta_t^{\text{lab}} - \theta_{t-1}^{\text{lab}}$, so $V_u = \sigma_{\psi}^2 + 2(1 - \rho_g)\sigma_{\theta}^2$, $V_l = 2\rho_g\sigma_{\theta}^2$, and $V_u + V_l = \sigma_{\psi}^2 + 2\sigma_{\theta}^2$.

The informative part of a pay change is u_t : were the worker to observe u_t^2 alone and re-optimize her gain for that cleaner signal, she would set it to σ_{ψ}^2/V_u and be unbiased, $\mathbb{E}[\hat{\sigma}_{\psi}^2] = \sigma_{\psi}^2$. That is the no-attenuation benchmark, not the attribution she carries. The rule of Section 5 updates on the full observed change $(u + l)^2$, so the attribution identified by (4) is the benchmark gain attenuated by the labelled part l that rides in that change, $\lambda_g = (\sigma_{\psi}^2/V_u) \text{Cov}(u^2, (u+l)^2)/\text{Var}((u+l)^2)$. Under joint Gaussianity with $u \perp l$, $\text{Cov}(u^2, (u+l)^2) = 2V_u^2$ and $\text{Var}((u+l)^2) = 2(V_u + V_l)^2$, so

$$\lambda_g = \frac{\sigma_{\psi}^2 V_u}{(V_u + V_l)^2} = \frac{\sigma_{\psi}^2 (\sigma_{\psi}^2 + 2(1 - \rho_g)\sigma_{\theta}^2)}{(\sigma_{\psi}^2 + 2\sigma_{\theta}^2)^2},$$

which is (7).

D Belief estimates and robustness

Table 7 reports the headline joint-GMM estimates of (4) plotted in Figure 4. The rest of this appendix checks their robustness: D.1 adds an occupation $\times$ time fixed effect to the shock con-

struction, D.2 varies the instrument lags, the trim, and the COVID handling one at a time, and D.3 reports the sign-split shock.

Group	σ_θ^2 (SIPP truth)	$\sigma_{\theta,\text{perc},g}^2$ (GMM)	λ_g (GMM)	μ_g (GMM)	Hansen J p	N
Bot	2.52	9.75 (1.71)	+ 0.077 (0.030)	0.628 (0.055)	0.20	1,531
Mid	4.42	2.94 (0.93)	+0.046 (0.015)	0.912 (0.089)	0.23	7,501
Top	5.83	4.12 (0.82)	+0.026 (0.013)	0.950 (0.060)	0.24	11,331
Wald ($B = T$)	—	$p = \mathbf{0.003}$	$p = 0.12$	$p = \mathbf{0.0001}$	—	—

Table 7: Joint Blundell-Bond system GMM on (4), by income group (the estimates of Figure 4). Two-way clustered SEs (respondent, quarter) in parentheses. Instruments: $\bar{\sigma}_{\text{perc}}^2$ at lags 3–4, the shock at lags 1–3, plus BB level moments. The Hansen J never rejects.

D.1 Occupation $\times$ time fixed effect

Adding an occupation $\times$ year-month fixed effect to the wage-shock construction (the interaction is the active term, since an additive occupation effect cancels in the within-spell difference) preserves the persistence and attribution gradients and sharpens the attribution; the prior gradient holds in direction but loses cross-group significance, so part of the bottom’s inflated prior reflects cross-occupation transitory dispersion. The labelling channel behind λ is robust to occupation controls; the reading-noise channel behind the prior operates partly through cross-occupation variation.

Group	Canonical			Occupation $\times$ time FE		
	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta,\text{perc},g}^2$	$\hat{\mu}_g$	$\hat{\lambda}_g$	$\hat{\sigma}_{\theta,\text{perc},g}^2$
Bot	0.628 (0.055)	+ 0.077 (0.030)	9.75 (1.71)	0.649 (0.063)	+ 0.091 (0.039)	7.20 (2.66)
Mid	0.912 (0.089)	+0.046 (0.015)	2.94 (0.93)	0.924 (0.091)	+0.036 (0.010)	2.74 (0.96)
Top	0.950 (0.060)	+0.026 (0.013)	4.12 (0.82)	0.957 (0.061)	+0.017 (0.008)	4.17 (0.74)
Wald μ (T–B)	$p = \mathbf{0.0001}$			$p = \mathbf{0.0004}$		
Wald λ (B–T)	$p = 0.118$			$p = 0.062$		
Wald σ^2 (B–T)	$p = \mathbf{0.003}$			$p = 0.274$		

Table 8: Canonical specification versus the same regression with an occupation $\times$ year-month fixed effect in the wage-shock construction. Standard errors in parentheses.

D.2 Lag depth, trim, and COVID exclusion

Each row of Table 9 varies one element of the canonical specification. The persistence and attribution gradients are robust across the lag variants, with the attribution Wald tighter than canonical in three of four. The within-group trim preserves the qualitative pattern but loosens both Walds. Excluding COVID sharpens the attribution gradient and breaks the prior gradient’s significance, so part of the bottom’s elevated prior is identified from the COVID year. The Hansen J never rejects.

The bottom series drifts upward over the sample (+0.83 per year, $t = 3.7$), which strains the mean stationarity behind the Blundell-Bond level moments that identify the constant, and through it the prior. The prior does not rest on them: removing a linear calendar trend or calendar-year effects before estimation leaves it essentially unchanged (9.74 and 9.57 against 9.65), dropping the level instruments entirely leaves it at 9.22 (1.81) while costing precision only on the gain, and the

split-sample estimates of 6.08 and 10.46 remain well above the realised 2.52.

Variant	$\hat{\mu}_g$			$\hat{\lambda}_g$			$\hat{\sigma}_{\theta, \text{perc}, g}^2$			Wald p (B vs T)		
	B	M	T	B	M	T	B	M	T	μ	λ	σ^2
Canonical	0.628	0.912	0.950	+0.077	+0.046	+0.026	9.75	2.94	4.12	0.0001	0.118	0.003
Trim 2.5/97.5	0.816	0.847	0.897	+0.059	+0.053	+0.014	11.23	4.07	4.86	0.110	0.352	0.000
$\bar{\sigma}^2$ lags 2–3	0.648	0.951	0.876	+0.075	+0.050	+0.020	9.96	2.85	4.44	0.000	0.079	0.003
$\bar{\sigma}^2$ lags 4–5	0.588	0.901	0.948	+0.102	+0.046	+0.021	9.03	3.00	4.38	0.000	0.061	0.032
Shock lags 0–1	0.683	0.937	0.940	+0.082	+0.016	+0.012	9.76	4.41	4.90	0.015	0.063	0.010
Shock lags 2–4	0.675	0.908	0.948	+0.095	+0.040	+0.027	8.76	3.20	4.10	0.003	0.008	0.011
Exclude 2020–21	0.636	0.995	0.886	+0.139	+0.043	+0.013	6.50	3.12	4.88	0.017	0.055	0.612

Table 9: One-at-a-time variations around the canonical specification. Bold p -values reject cross-group equality at the 5% level.

D.3 Asymmetric extrapolation

Pooling across groups and splitting the shock by sign, both components load positively, with positive news carrying roughly twice the weight of negative news ($\lambda^+ = 0.055$ versus $\lambda^- = 0.027$), significant across cluster choices ($N = 20,363$, Hansen J $p = 0.11$). Upside news is read as informative about future earning power; downside news is more often dismissed as transitory, the opposite of loss aversion.

D.4 Numeracy and inflation uncertainty

The SCE’s numeracy module classifies 47.0 percent of bottom-group respondents as low-numeracy, against 28.4 percent in the middle and 11.1 at the top, so any mechanical link between numeracy and the width of a reported density is graded the same way as the headline. Conditioning on it compresses the level gradient without removing it: on the full matched panel, a broader sample than the estimation panel, the bottom-to-top difference in perceived variance is 10.6 (1.0), falling to 2.4 (0.8, $p = 0.002$) among high-numeracy respondents and rising to 17.0 (2.1) among low-numeracy ones, with high-numeracy group means of 10.7/8.6/8.7. Re-estimating the belief block (4) within high-numeracy respondents reproduces the middle and the top, with priors of 2.35 (0.85) and 4.93 (1.02), at or below their realised values as in the baseline; the bottom retains only 75 estimation observations under the numeracy-module overlap and its cell is degenerate, so whether the bottom’s prior excess survives the conditioning cannot be answered at the estimation level. Two readings of the compression are open: numeracy could widen reported densities mechanically, making the conditioned gap the cleaner estimate, or numeracy is part of the capacity to read fragmented pay, in which case conditioning removes part of the reading-noise channel itself and the conditioned gap is a lower bound, the same over-control argument as the occupation effect of D.1. The gradient survives either reading; its size is bracketed by the two.

Inflation uncertainty is the parallel confound: it is itself steeply graded, 27.9/15.2/8.5 across the groups, and controlling for it absorbs about seventy percent of the bottom-to-top wage-risk gap, leaving a wage-specific gradient of +3.09 (0.60) that remains far from zero. The same two readings apply: general uncertainty could widen the reported wage density mechanically, making the residual the cleaner estimate, or the environment that makes pay hard to read makes prices hard to read

too, in which case the control over-partials again. A wage-specific gradient survives either reading.

D.5 Persistence of the realised volatility process

A rational agent tracking a conditional variance that is itself more persistent at the bottom would show the belief-persistence gradient of Table 1 with no belief friction at all. The realised process does not deliver that: on the raw, unsmoothed cell series underlying the shock regressor, the within-cell monthly AR(1) of the squared change is 0.29 at the bottom against 0.36 at the top (0.44 against 0.49 in the national series), and instrumenting the lag to remove finite-cell sampling noise drives the bottom to 0.06 against the top's 0.26; the bottom's quarterly national series is mildly anti-persistent. The mechanical correlation that adjacent squared changes share through the same person-month is larger where cells are smaller, at the bottom, so the uninstrumented numbers overstate the bottom if anything. The greater persistence of the bottom's beliefs is therefore not inherited from its volatility process.

E Hours by income group

Both forces operate through hours: fragmented, variable hours generate the reading-noise force that lifts the bottom's prior, and transitory shocks delivered through hours are of the unlabelled kind and lower ρ_g . Within-person hours volatility $\mathbb{E}[(\Delta \log h)^2]$ falls monotonically with income (4.10/3.06/2.36 in $(\%)^2$), consistent with the reading-noise force being highest at the bottom (Figure 8). The hours proxy is a one-directional check on it: it supports a reading-noise channel that falls with income, the gradient of the residual above the netting floor (Section 7), while the priors higher up sit on their measured floors (Appendix F). The gradient is directional only: a proportional scaling of the residual in hours volatility, calibrated at the bottom (7.43/4.10), would predict a middle residual near 5.5 against an estimated -0.68 (standard error 0.94), so no intensity mapping from observables to ω^2 is estimable from three group means, two of them zero.

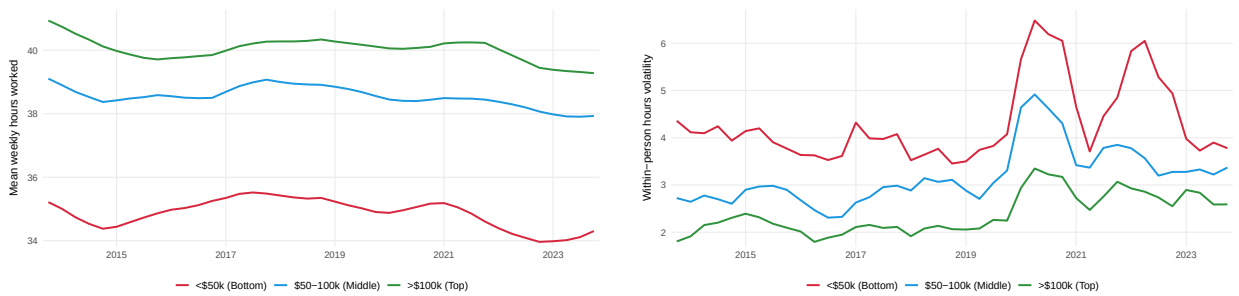


Figure 8: Mean weekly hours (left) and within-person hours volatility $\mathbb{E}[(\Delta \log h)^2 | g, t]$ in $(\%)^2$ (right) by income group. SIPP matched-stayer panel, quarterly with 4-quarter trailing MA.

F The filter, the measurement of the labelled share, and the scorecard

This appendix gives the filter behind the gain, details the measurement of the labelled share and its robustness, reports the scorecard that sets the measured mechanism against the estimated belief moments, and defines the upper-bound belief-anchored variant that Section 10 reports.

The filter. The agent’s monthly signal is one squared pay reading, $z_t = (\Delta \tilde{y}_t)^2$ with $\tilde{y}$ the netted reading of (5), with mean $\sigma_\psi^2 + V_{x,g}$ and, under Gaussianity, variance $2(\sigma_\psi^2 + V_{x,g})^2$, where $V_{x,g} = 2\sigma_{\theta,\text{perc},g}^2$ is twice the perceived transitory prior. Treating the permanent variance as a random walk with drift q observed through the attributed signal $\lambda_g z_t$, the steady-state Kalman gain is

$$\mu_g = \frac{\bar{P}_g}{\bar{P}_g + R_g}, \quad R_g = \lambda_g^2 2(\sigma_\psi^2 + V_{x,g})^2, \quad \bar{P}_g = \frac{1}{2}(q + \sqrt{q^2 + 4qR_g}).$$

The attribution applied to this signal is the full-change attribution (7) rescaled to the signal’s own mean, $\lambda_g^{\text{rule}} = \sigma_\psi^2 V_u / [\bar{S}_m(\sigma_\psi^2 + V_{x,g})]$, so that the filter’s resting mean is the permanent belief (8); substituting it collapses the observation noise to $R_g = 2(\sigma_\psi^2)^2 (V_u / \bar{S}_m)^2$, so the gain depends on the labelled share and not on reading noise, the perceived prior entering the attribution and the signal variance by offsetting factors and cancelling out of the signal-to-noise ratio. The model carries the gain as the estimated moment; the filter serves as the diagnostic on its ordering reported below.

Measurement of the labelled share. The SIPP records, for each job and month, the dollar amounts of bonuses, commissions, tips, and overtime pay alongside total earnings. On the stayer panel of Section 2, same-employer spells of at least six consecutive months over reference years 2013–2023, 2,250,815 person-months across 103,134 persons and 127,184 spells, I demean each component within its spell, scale the deviations by spell-mean earnings, and estimate ρ_g as the regression share of the labelled components in the total within-spell deviation, the covariance of the labelled deviation with the total over the variance of the total, weighted by the SIPP person weights. The main specification labels bonuses and commissions, the components that arrive flagged on their own line, and leaves tips and overtime in the unlabelled residual, tips because they arrive as scattered cash, overtime because a variable shift count is exactly the illegible delivery of Section 7. Seam months are dropped as in Section 2. Table 10 collects the specifications.

Specification	Bot	Mid	Top
Main: within-spell demeaned, labelled = bonus + commission	0.121	0.178	0.358
Variance share instead of regression share	0.146	0.203	0.373
Within-spell detrended	0.176	0.256	0.488
Imputed component months dropped	0.042	0.085	0.314
Overtime counted as labelled	0.134	0.197	0.370
Annual spell-year cells (different frequency)	0.050	0.060	0.075

Table 10: The measured labelled share ρ_g across specifications, by income group, SIPP stayer panel 2013–2023, person-weighted. The main row is the specification the model carries. The detrended row removes a within-spell linear trend that demeaning would otherwise leave in the unlabelled residual, and bounds the share from above; the imputation row drops person-months in which any pay component is imputed, and bounds it from below. The annual row aggregates to spell-year cells before computing shares and is reported for completeness, not comparability.

Two caveats attach. First, component amounts in the SIPP are partly imputed, and imputation that spreads labelled dollars across months can manufacture labelled variance; dropping every imputed person-month lowers the shares to 0.042/0.085/0.314 but leaves the gradient intact, so I treat that row as the lower edge of the band rather than the preferred estimate, since it also discards genuine labelled months. Second, the annual specification is a different-frequency object:

aggregated to spell-year cells, the month-to-month variation the filter operates on has largely averaged out, and near-annual components such as bonuses contribute little variance within a year, so the annual share measures the labelled fraction of year-to-year risk, not of the monthly signal the belief recursion reads, and it does not enter the band.

The scorecard. With ρ_g measured, the estimated belief moments of Section 6 become out-of-sample predictions, and four comparisons score them. First, the netting floors $(1 - \rho_g)\sigma_\theta^2 = 2.21/3.63/3.74$ place the middle and top priors on their floors: the residuals $\sigma_{\theta, \text{perc}, g}^2 - (1 - \rho_g)\sigma_\theta^2 = \hat{\omega}_g^2$ of (6) are +7.43, -0.68, and +0.38, with standard errors 1.76, 0.94, and 0.82, so the bottom's residual is the only one that differs from zero ($t = 4.2$), and it is the reading-noise force, matching the hours-volatility gradient 4.10/3.06/2.36 of Appendix E; at the reconciled shares the model carries (Section 9) the floors are 2.22/3.55/3.68 and the residuals +7.43/-0.59/+0.44, the same picture. Second, netting predicts the middle and top priors to fall short of realised transitory risk by $\rho_g\sigma_\theta^2 = 0.79$ and 2.09, against estimated shortfalls of 1.46 and 1.71; the gaps are $z = -0.7$ and +0.5, each within one standard error. Third, the closed form (7) at the measured shares predicts bottom-to-top and middle-to-top attribution ratios of 1.69 and 1.31 (in logs 0.52 and 0.27), against estimated ratios of 3.04 and 1.76 (in logs 1.11 and 0.57, with log standard errors 0.63 and 0.59); the gaps are $z = +0.9$ and +0.5, within one standard error, and the predictions are about half of the estimated point gradient. Fourth, the gain diagnostic below predicts the ordering of the estimated gains and underpredicts their spread. Three of the four moments are hit within one standard error; the one miss lands on the gain levels, the moments the cadence comparison of Section 3 marks as the least reliable in the belief block.

Attribution: ratios, not the level. The cell-mean regressor attenuates the estimated attribution: the cell mean tracks only the common component of a worker's shock, so the slope is scaled down by the share of the squared-shock variance that is common. Treated as common across groups, the factor cancels in the cross-group attribution ratios, which are the moments the scorecard compares; the level is not comparable. The common-factor restriction is an identifying assumption of the merge, not something the data verify. The structural attribution at the measured shares, 0.677/0.524/0.401, stands against the regression level 0.079/0.046/0.026, an implied attenuation of roughly an order of magnitude; that the implied factor is not quite uniform across groups is the same statement as the scorecard's half-explained ratio gradient. The two-percent between-cell share of the squared shock in SIPP makes an attenuation of this order plausible, without pinning it.

The gain diagnostic. The gain is carried, not derived, but the filter above disciplines its ordering. At the measured shares the observation noise is $R_g = 2(\sigma_\psi^2)^2(V_u/\bar{S}_m)^2 = 262.7/229.2/168.3$, falling with the labelled share, so any common drift q implies a gain rising with income, which is the estimated ordering. At the drift that comes closest to the three estimated gains, $q^* = 646.5$, the predicted gains are 0.763/0.783/0.823 against the estimated 0.629/0.910/0.950: the ordering is right and the spread underpredicted. The diagnostic carries no weight anywhere in the calibration, since the model reads the estimated gains directly, and the gain levels are in any case the moments Section 3 shows the two survey cadences cannot jointly pin.

The upper-bound variant. Section 10 reports a variant that takes the estimated attribution responses at face value instead of the measured delivery. It anchors the bottom's attribution at

its full-information share $\lambda^* = \sigma_\psi^2 / \bar{S}_m = 0.702$ and lets the estimated attribution ratios supply the middle and the top, giving $\lambda = 0.702/0.409/0.231$; the resting permanent belief $\lambda \bar{S}_m$ is then 11.89/8.36/5.29, discounts of 1.00/0.72/0.47 on the truth against 0.96/0.92/0.82 under the measured shares, with the prior carried as in the baseline. The variant is an upper bound on the mechanism: it attributes the whole of the estimated attribution gradient to delivery-graded under-reaction, so the gap between its wedge and the measured calibration’s measures what rides on netting beyond the measured pay components, or on sampling noise in the belief estimates.

Why three groups. The continuous profiles of Figure 5 interpolate three income groups; finer bins cannot anchor the belief moments. The SCE elicits income in brackets whose extremes are thin, and splitting the terciles into the five native brackets and re-estimating breaks the belief block exactly where it is split: the lowest bracket (under \$20k, 102 respondents) retains no cell that clears the estimation floor, the attribution turns negative in the lower-middle bracket, and the gain exceeds one (an inadmissible negative persistence) in the top two brackets; only the middle bracket, which is the unchanged tercile, survives. System GMM with lagged instruments and two-way clustering needs hundreds of within-person spells per cell, and the bracketed extremes do not supply them. The terciles are the finest split the survey powers, and the continuum is an interpolation through them, not an estimate.

Interpolation. The income profiles in Figure 5 interpolate the three groups with monotone cubic interpolants, the measured share and the carried gain in levels and the carried prior in logs, each held flat outside the anchor incomes so that no belief object is extrapolated beyond its measured range; the tangents at the end anchors are set to zero, so each profile joins its flat extension smoothly rather than at a kink, and the interior tangent carries the Fritsch–Carlson bound that keeps every segment monotone. The prior’s interpolant is additionally censored below at the local netting floor $(1 - \rho(y)) \sigma_\theta^2(y)$, since the mechanism admits no prior beneath it; with the flat tangent at the middle anchor the constraint never binds between anchors, the carried curve touching its floor only at the middle itself. The attribution, the discount, and the permanent belief follow from the closed forms at each income. With three anchors the interpolants set only the shape between groups. Table 11 evaluates the resulting mapping at representative incomes.

Monthly	Measured	Carried		Attribution		Discount	Perceived
income	ρ	μ	$\sigma_{\theta, \text{perc}}^2$	λ	λ^{rule}	$V_u/\bar{S}_m$	perm. var
\$1,500	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$2,415 (Bot)	0.119	0.629	9.65	0.677	0.368	0.965	11.47
\$3,500	0.134	0.718	6.72	0.610	0.445	0.952	11.21
\$6,014 (Mid)	0.197	0.910	3.55	0.519	0.568	0.915	10.61
\$9,000	0.297	0.945	3.82	0.453	0.515	0.860	9.81
\$13,604 (Top)	0.369	0.950	4.12	0.398	0.468	0.812	9.11
\$20,000	0.369	0.950	4.12	0.398	0.468	0.812	9.11

Table 11: The income-to-belief mapping of Figure 5 evaluated at representative incomes. The reconciled share ρ and the carried gain μ are interpolated through the three group anchors (the Bot, Mid and Top rows) with monotone cubic interpolants in levels, and the carried prior $\sigma_{\theta, \text{perc}}^2$ in logs, all held flat outside the anchors with smooth joins, and the prior censored below at its netting floor between them (Appendix F; the constraint never binds). The attribution λ is the closed form (7) at each income; λ^{rule} is the recursion’s gain, the attribution rescaled to the agent’s own perceived signal so that the resting mean equals the perceived permanent variance; the discount $V_u/\bar{S}_m$ and the perceived permanent variance $\sigma_\psi^2(V_u/\bar{S}_m)$ follow from the same identities. Rows away from the three groups are interpolated, not separately measured. In $(\%)^2$ except the dimensionless shares.

G Steady-state model output

Table 12 gives the numbers behind Figure 6: the perceived variance under the measured beliefs against the full-information benchmark, by income group.

Group	$\hat{\sigma}_\psi^2$	$\sigma_{\theta, \text{perc}}^2$	Perceived $\bar{\sigma}_{\text{perc}}^2$	FIRE	Ratio
Bot	11.3	8.3	153	148	1.03
Mid	10.8	5.1	139	148	0.94
Top	9.7	3.9	124	147	0.84

Table 12: Stationary perceived variance against the FIRE benchmark, in $(\%)^2$ at the 12-month horizon, simulated cross-sectional means by income tercile. $\hat{\sigma}_\psi^2$ is the mean permanent-variance belief, against realised tercile means of 11.8/11.6/11.4, and $\sigma_{\theta, \text{perc}}^2$ the mean transitory prior, against realised 2.9/4.0/5.3, the crossing of Section 7 in the simulated cross-section; $\bar{\sigma}_{\text{perc}}^2 = 12\hat{\sigma}_\psi^2 + 2\sigma_{\theta, \text{perc}}^2$; FIRE = $12\sigma_\psi^2 + 2\sigma_\theta^2$. Tercile means mix types; the bottom tercile’s total sits mildly above FIRE, its prior excess outweighing a small permanent discount.

Table 13 gives the numbers behind Figure 7: mean liquid wealth and the 12-month MPC, belief regime versus FIRE.

Group	Mean wealth (months)			12-month MPC		
	FIRE	Belief	Δ	FIRE	Belief	Δ
Bot	2.29	2.25	-1.9%	0.562	0.564	+0.3pp
Mid	1.71	1.65	-3.4%	0.504	0.510	+0.6pp
Top	1.47	1.38	-6.3%	0.481	0.494	+1.3pp
<i>Workers only, re-terciled among workers</i>						
Bot	2.10	2.05	-2.2%	—	—	—
Mid	1.50	1.43	-4.1%	—	—	—
Top	1.40	1.30	-7.0%	—	—	—

Table 13: Belief regime versus FIRE, by income tercile, under the single calibrated discount factor and mobility. Wealth in months of permanent income; MPC over 12 months out of a small transitory windfall, common-random-numbers paired. PE, $r = 2$ percent, $\beta = 0.879$ common (calibrated to mean liquid wealth ≈ 1.8 months), so the wealth and MPC levels are in the data’s range and the wedge in Δ is the object of interest. The bottom current-income tercile is 54 percent retirees, the shares falling to 23 and 11 percent above; the workers-only panel restricts to working households, re-terciled among workers, the MPC not recomputed for the split.

Table 14 gives the general-equilibrium counterpart of Section 10: the FIRE and belief economies cleared at a common productivity normaliser pinned from FIRE at a three-percent target rate.

Regime	r (% ann.)	K	Wealth Gini	H2M share
FIRE	3.00	2.83	0.654	0.288
Belief	3.12	2.74	0.657	0.295

Table 14: General-equilibrium FIRE versus belief, common $\beta = 0.879$ (the calibrated value), Z pinned from FIRE at $r^* = 3\%$ and held fixed across regimes. The belief economy under-perceives permanent risk, supplies 3.4% less capital, and clears at a rate 0.12pp higher with a wealth Gini 0.003 higher and a slightly higher hand-to-mouth share. One-asset closure; a two-asset extension is left to future work.